Understanding Qiskit's SABRE implementation

SABRE is the algorithm Qiskit uses to make a circuit runnable on a real device. A device only lets certain pairs of physical qubits interact, so SABRE decides **where each program qubit should start** and **which SWAP gates to insert** so that every two-qubit gate lands on a connected pair. The name stands for **SWAP-based BidiRectional heuristic search**; the algorithm was introduced by Li, Ding, and Xie in Tackling the Qubit Mapping Problem for NISQ-Era Quantum Devices.

Who this is for. You know that a circuit is a list of gates acting on qubits, and that a CX gate touches two qubits. That is all you need. There is no state-vector mathematics, no graph theory beyond "neighbors", and no Rust.

What you will be able to do afterwards.

- Route a small circuit by hand, making exactly the choices SABRE makes.
- Compute the score of each candidate SWAP and say why one wins.
- Explain how routing the circuit *backward* produces a better starting layout.
- Find the code that performs each step.

What this guide describes. The Rust code in `crates/transpiler/src/passes/sabre` of **this checkout**. It is classic SABRE (distance, lookahead, decay) plus two extra scoring terms added here: a *fusion credit* and a *ready-time penalty*. Parts I, II and IV explain classic SABRE, and say so explicitly at the decisions where this checkout's extra terms change the outcome: once by overriding the lookahead (decision 3 of Part II) and three times by settling a tie that classic SABRE would leave to chance. Part III explains those terms.

How to read it.

If you want…	Read
The idea in ten minutes	Part I, especially section 3
To understand the algorithm properly	Parts I and II in order; every decision is worked in full
What this checkout changed	Part III
How the initial layout is chosen	Part IV
Code, runnable examples, FAQ	Part V

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After this guide, read SABRE_CHANGE_EXPLAINED.md for the motivation, benchmarks and limitations of the change made in this checkout.

Notation used everywhere in this guide

- `q1`, `q2`, … are **program qubits** (also called logical or virtual qubits); Part I's small example calls them `A`, `B`, `C`. `p0`, `p1`, `p2`, … are **physical positions** on the device. Positions never move; program qubits do.
- A row in brackets lists **who sits where**, position by position. `[q1, q2, q6, q3, q4, q5]` means q1 is at p0, q2 at p1, q6 at p2, q3 at p3, and so on.
- `SWAP(p0, p1)` exchanges the occupants of positions p0 and p1.
- `d(q2, q3)` is the number of hardware edges between the positions that currently hold q2 and q3. Neighbors have distance 1.
- Scores: **smaller is better**. A negative score change means "this SWAP makes things better"; it is not a negative number of gates.
- In the state figures, an edge label such as `q2,p1` means "this dependency follows q2, which is currently at p1". A dashed line below a gate marks the **front layer F**; *** below a gate marks the **lookahead layer E**. Grey gates are done; green gates can be emitted next. Colour is reinforced by text and symbols.
- Each state figure includes the current layout and the **logical-qubit distance matrix**. The hardware distance matrix stays fixed; the logical matrix follows the qubits as they move. A cell of 1 means adjacent, but the gate still needs its predecessors to be done.

Part I — The problem

1. The problem in one picture

Imagine three people, A, B and C, sitting in a row of three chairs. They may only whisper to the person in the chair next to them, and they may trade chairs with a neighbor, but every trade takes time. The program says "A must whisper to C". They are two chairs apart, so somebody has to move.

The quantum version is the same picture:

```
Physical positions:   p0 ---- p1 ---- p2
Occupants:             A       B       C

Program:  CX(A, C)   then   CX(A, B)
```

The device only connects p0–p1 and p1–p2. The first gate needs A and C to be neighbors, and they are not. There are two ways to fix it with one SWAP:

```
Option 1: SWAP(p0, p1)    ->   [B, A, C]     A and C are now at p1, p2: adjacent.
Option 2: SWAP(p1, p2)    ->   [A, C, B]     A and C are now at p0, p1: adjacent.
```

Both options make the first gate possible. They are **not** equally good, because of the *second* gate, `CX(A, B)`:

After	Where A and B are	d(A, B)	Extra SWAP needed for CX(A, B)?
Option 1, `[B, A, C]`	p1 and p0	1	No
Option 2, `[A, C, B]`	p0 and p2	2	Yes, one more

Option 1 solves the whole program with one SWAP; option 2 needs two. Choosing between such options, quickly and reasonably well, for circuits with thousands of gates, is SABRE's entire job.

There are two decisions to make, and SABRE makes both:

Decision	Question	In the example
Initial layout	Where should each program qubit *start*?	If A had started at p1 and C at p2, no SWAP would be needed at all.
Routing	Given where everyone is now, which SWAPs make the next gates possible?	Insert SWAP(p0, p1).

Choosing the initial layout costs no gates on the device: it is pure compile-time search (Part IV) that assigns program qubits to positions before anything runs. Routing is not free: every SWAP is a real operation on the device. On most

hardware a SWAP is built from **three** CX gates, so each SWAP adds noise and time. SABRE therefore tries to use few SWAPs and, secondarily, to let independent SWAPs run at the same time.

2. The vocabulary you need

Gates and SWAPs

Term	What it means for routing
One-qubit gate	Acts on one qubit. Never needs routing: a single qubit is always "connected to itself".
Two-qubit gate	Acts on two qubits. The two must sit on **connected** physical positions.
`CX` (CNOT)	The most common two-qubit gate. The first operand is the control, the second the target; the order matters for the gate's meaning.
`SWAP`	Exchanges the quantum states held by two positions. The hardware positions stay put; the information moves. Usually implemented as three CX gates.
Measurement, barrier	Not routing constraints themselves, but they are ordering points that other gates must respect.

A SWAP does not copy or measure anything. After `SWAP(p0, p1)`, whatever A's state was is now at p1, and B's state is at p0. The compiler must remember this, which is what the layout does.

Program qubits, physical positions, and the layout

A **program qubit** (logical or virtual qubit) is a name in the user's circuit: A, B, C here, q1 … q6 from Part II on. A **physical position** (physical qubit) is a numbered place on the device: p0, p1, p2, … The **layout** is the table that says who is where. It changes every time a SWAP is inserted:

Program qubit	Position at start	Position after SWAP(p0, p1)
A	p0	p1
B	p1	p0
C	p2	p2

Every later gate must be placed using the *current* layout. "CX(A, C)" is always a gate on A and C; which physical positions that means depends on when it runs. In the Rust code the layout is an `NLayout`, which stores both directions (`layout[A]` gives A's position, and the reverse lookup gives the occupant of a position) and has a `swap_physical(a, b)` method.

The device: coupling map and distance

The **coupling map** is the device's connection graph: positions are nodes, and an edge means "a two-qubit gate can run on this pair". The **distance** between two positions is the smallest number of edges on a path between them. For the four-position line used in this guide:

```
p0 ---- p1 ---- p2 ---- p3
```

From / to	p0	p1	p2	p3
p0	0	1	2	3
p1	1	0	1	2
p2	2	1	0	1
p3	3	2	1	0

This table is computed once from the coupling map and never changes. The target for a two-qubit gate is distance **1**, not 0: two different qubits must be in two different positions. A pair at distance `d` needs at least `d - 1` SWAPs to become adjacent if we move it along a shortest path, but one SWAP can also help two pairs at once, or hurt one while helping another.

The circuit as a dependency graph

Two gates that share a qubit must run in the order the program wrote them. Gates on disjoint qubits are independent and may run in either order or at the same time. This "must come before" relation forms a **DAG** (directed acyclic graph), and SABRE follows it strictly: it never reorders two gates that share a qubit.

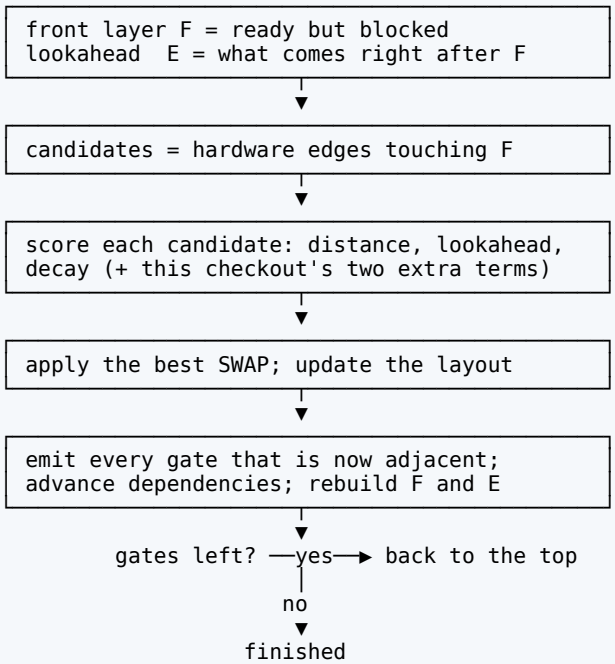
What "cost" means

Measure	Meaning
SWAP count	Number of SWAPs SABRE inserted. This is what SABRE itself uses to compare complete attempts.
Two-qubit gate count	Number of **native** two-qubit gates, meaning gates the device can run directly, after later compiler passes have translated everything into them. Each SWAP is usually 3 native gates, sometimes fewer after optimization.
Two-qubit depth	Number of sequential layers of two-qubit gates. Independent gates can overlap, so depth is not the same as count.

3. SABRE in one page

Here is the whole routing algorithm in plain words. Everything after this section is detail.

1. **Find the blocked gates.** Look at the gates whose predecessors are all done but whose two qubits are not adjacent. This set is the **front layer**. (Gates that are ready *and* adjacent are simply **emitted** at once: written to the output circuit on their current physical positions.)
2. **Peek ahead.** Find the gates that would become ready right after the front layer. This is the **lookahead layer**. It is only looked at, never executed early.
3. **List the candidate SWAPs.** Every hardware edge that touches a qubit of a front-layer gate is a candidate. Nothing else is.
4. **Score each candidate.** Pretend to apply it, and measure how much the front-layer distances shrink. Add a small fraction of how much the lookahead distances shrink. (This checkout adds two more small terms, explained in Part III.) Smaller is better.
5. **Apply the best candidate.** Insert that SWAP into the output and update the layout. If several candidates tie exactly, pick one at random.
6. **Emit what is now possible.** Any front gate that became adjacent is emitted. That may unlock successors; emit those too if they are adjacent, otherwise they join the front layer.
7. **Repeat** from step 1 until no gates remain.



The smallest complete example

Take the three-chair program from section 1 and run the loop once.

```
Hardware:   p0 ---- p1 ---- p2         Occupants: [A, B, C]
Program:    g1 = CX(A, C),  g2 = CX(A, B)
```

Step 1, front layer. g1 is ready (nothing precedes it) but A is at p0 and C at p2, distance 2. So `F = {g1}`. g2 is not ready: it must wait for g1 because they share A.

Step 2, lookahead. If g1 were done, g2 would be ready. So `E = {g2}`.

Small example / steps 1-4

Read the front, build lookahead, enumerate candidates, score. The real state stays fixed.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	A	B	C
A	0	1	2
B	1	0	1
C	2	1	0

Layers at this snapshot

F = {g1}

E = {g2}

Gate / physical operands / distance

g1: (p0,p2) d = 2

g2: (p0,p1) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 1: Steps 1-4 share this state: g1 is in F, g2 is in E, and the dependency edge follows A at p0. The matrix gives d(A,C)=2 and d(A,B)=1.

Read the matrix by **logical qubit name**, in the fixed order A, B, C. The row A, column C entry is 2. The layout strip above the graph tells you why: A is at p0, C at p2. The highlighted matrix cells are the pairs scored in F (blue) and E (amber); symmetric cells repeat the same distance.

Step 3, candidates. The front qubits sit at p0 and p2. Edges touching them: `(p0, p1)` and `(p1, p2)`. Two candidates.

Step 4, score. For each candidate, imagine applying it and measure the distances:

Candidate	Imagined occupants	d(A, C) front	change	d(A, B) lookahead	change
none (now)	[A, B, C]	2	—	1	—
SWAP(p0, p1)	[B, A, C]	1	−1	1	0
SWAP(p1, p2)	[A, C, B]	1	−1	2	+1

Both candidates improve the front gate by the same amount, so the front distance alone cannot choose. The lookahead gate breaks the tie: SWAP(p1, p2) would pull A and B apart. With the lookahead weight used for a three-qubit device, `w = 0.5 / 3 ≈ 0.167`:

```
score(p0, p1) = −1 + 0.167 × 0   = −1.000   <- best
score(p1, p2) = −1 + 0.167 × (+1) = −0.833
```

Step 5, apply. Insert `SWAP(p0, p1)`. The occupants become `[B, A, C]`.

Small example / step 5

Apply SWAP(p0,p1). g1 becomes executable; dependencies have not changed.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	A	B	C
A	0	1	1
B	1	0	2
C	1	2	0

Layers at this snapshot

F = {g1}

E = {g2}

Gate / physical operands / distance

g1: (p1,p2) d = 1

g2: (p1,p0) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

Inside the emit sweep: the marked F/E are not yet a new scoring state. Green means executable.

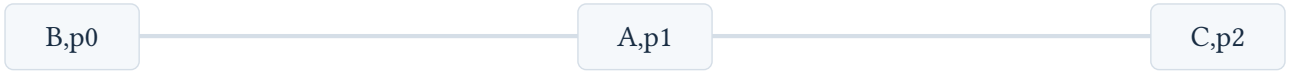
Figure 2: Step 5: the edge label changes to A,p1 and d(A,C) becomes 1. g1 can now be emitted, but the SWAP itself has completed neither program gate.

Step 6, emit. g1 = CX(A, C) is now on p1 and p2, adjacent: emit it. That unlocks g2 = CX(A, B), on p1 and p0, adjacent: emit it too.

Small example / step 6, after g1

Emit g1 on (p1,p2). g2 is now ready and adjacent: emit it on (p1,p0).

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	A	B	C
A	0	1	1
B	1	0	2
C	1	2	0

Layers at this snapshot

F = {}

E: rebuild pending

Gate / physical operands / distance

g2: (p1,p0) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

Inside the emit sweep: the marked F/E are not yet a new scoring state. Green means executable.

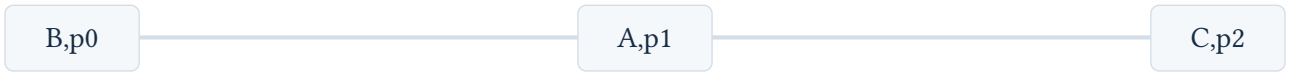
Figure 3: Inside step 6, after g1: g1 is done and g2 is executable. The matrix is unchanged because emitting a CX does not move logical qubits.

Step 7. Nothing is left. Routing is finished with one SWAP:

Small example / step 7

Both gates emitted. The final layout is A at p1, B at p0, C at p2.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	A	B	C
A	0	1	1
B	1	0	2
C	1	2	0

Layers at this snapshot

F = {}

E = {}

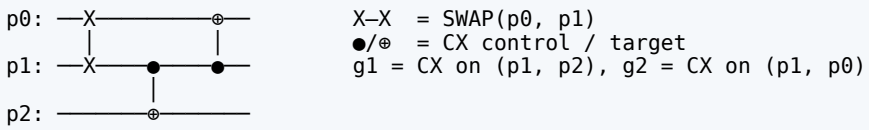
Gate / physical operands / distance

No gates remain to route.

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 4: Step 7: both gates are done, F and E are empty, and the final layout and distance matrix remain those of step 5.



Running this in Qiskit (section 21) produces exactly this circuit.

That is the whole algorithm. The remaining parts add: a slightly larger example where every decision is tabulated (Part II), the extra terms in the score (Part III), and how the same routine is reused to choose the initial layout by running it backward (Part IV).

Part II — Routing, one decision at a time

4. The running example

Everything in Parts II–IV uses one circuit on one device. The circuit is the one the SABRE paper uses to introduce the dependency graph and the front layer (Figure 4 of Li, Ding and Xie):

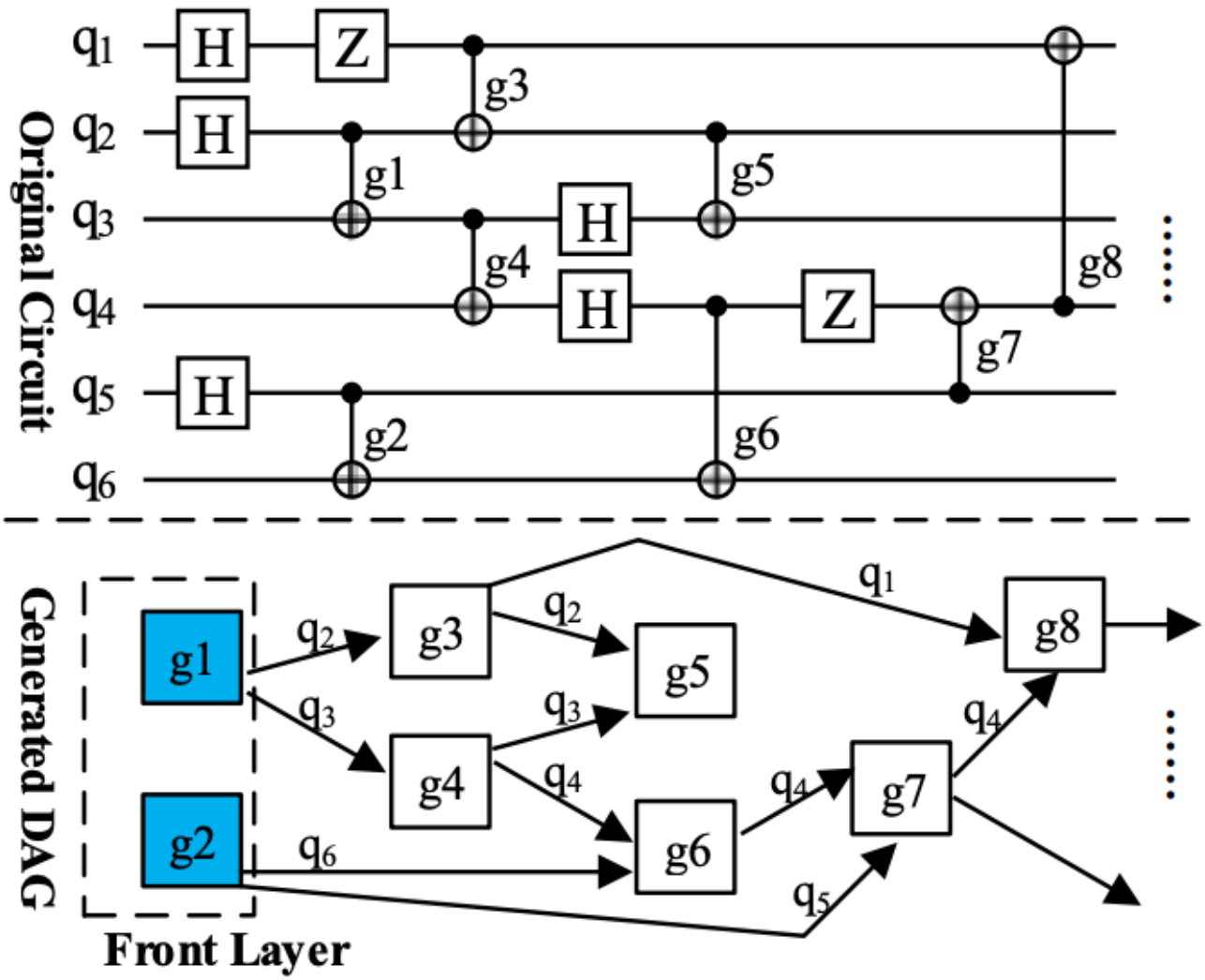
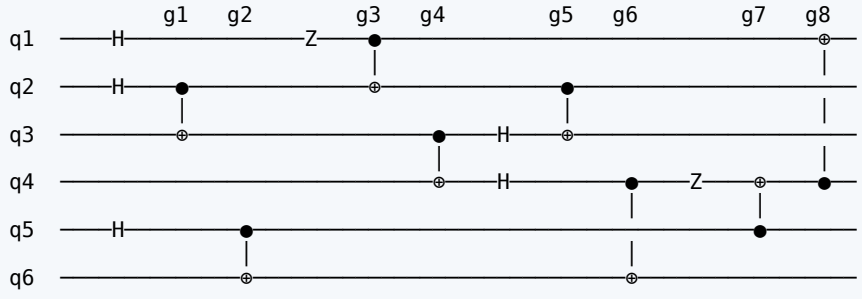


Fig. 4: Example of DAG Generation and Front Layer Initialization.

Figure 5: Figure 4 of the SABRE paper: the example circuit (top) and the dependency graph generated from it (bottom), with the initial front layer highlighted.

Six program qubits `q1` … `q6`, eight CX gates `g1` … `g8`, and seven one-qubit gates. The paper's drawing trails off after `g8`; we stop there. Redrawn with this guide's symbols (● = control, ⊕ = target):



Gate	Operation	One-qubit gates around it
g1	CX(q2, q3)	H on q1, q2 and q5 come first
g2	CX(q5, q6)	
g3	CX(q1, q2)	Z on q1 just before it
g4	CX(q3, q4)	H on q3 and H on q4 just after it
g5	CX(q2, q3)	
g6	CX(q4, q6)	Z on q4 just after it
g7	CX(q5, q4)	control q5, target q4
g8	CX(q4, q1)	control q4, target q1

One-qubit gates never need routing: whatever position a qubit occupies, a gate on that qubit alone can run there. Section 18 shows how the router folds them out of the way; until then the dependency graph only needs the CX gates.

Which gates must wait for which? Two gates are ordered only if they share a qubit:

Gate	Must wait for	Because
g1 = CX(q2, q3)	nothing	It is the first CX on both q2 and q3.
g2 = CX(q5, q6)	nothing	It is the first CX on both q5 and q6.
g3 = CX(q1, q2)	g1	They share q2. (q1 has no earlier CX.)
g4 = CX(q3, q4)	g1	They share q3.
g5 = CX(q2, q3)	g3 **and** g4	q2 comes from g3, q3 comes from g4.
g6 = CX(q4, q6)	g4 **and** g2	q4 comes from g4, q6 comes from g2.
g7 = CX(q5, q4)	g2 **and** g6	q5 comes from g2, q4 comes from g6.
g8 = CX(q4, q1)	g7 **and** g3	q4 comes from g7, q1 comes from g3.

As a dependency graph (arrows mean "must come before"). Each edge shows the shared logical qubit **and its physical location in the starting layout below**. For example, `g1 -> g3` carries `q2,p1`. The dashed underlines and asterisks mark F and E, introduced in section 5.

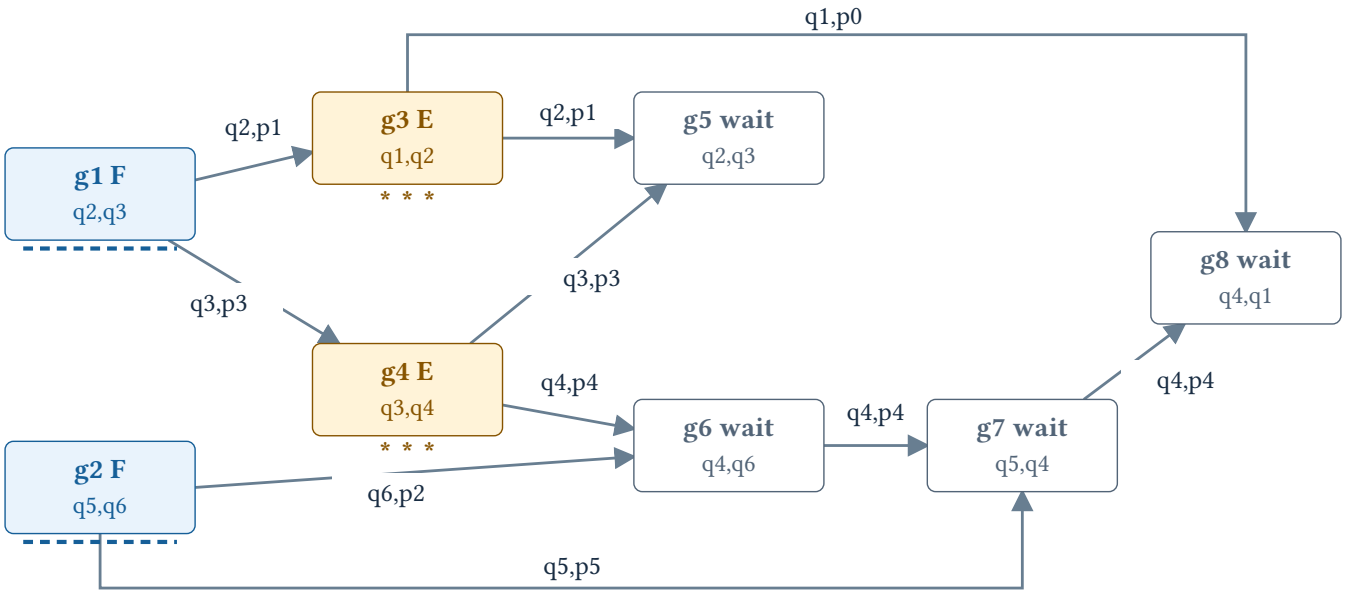


Figure 6: Initial dependency graph: every edge is labelled with its logical qubit and current physical location. g1 and g2 have front-layer underlines; g3 and g4 have lookahead asterisks.

Note that g1 and g2 are written one after the other but act on different qubits, so there is **no** arrow between them. Likewise g3 and g4: either may run first.

The device is a line of six positions:

```
p0 ---- p1 ---- p2 ---- p3 ---- p4 ---- p5
```

Starting occupants: `[q1, q2, q6, q3, q4, q5]`. Every program qubit sits at the position with its own number minus one, except q6, which has been parked between q2 and q3. (The paper leaves the device and the starting layout open. This choice makes both front-layer gates blocked at the start, so every part of the algorithm gets exercised. Part IV shows how SABRE would improve on it.)

The **physical distance matrix** for this device is fixed for the entire guide. Its row and column names are positions, so `D_phys[pi,pj] = |i-j|`:

From / to	p0	p1	p2	p3	p4	p5
p0	0	1	2	3	4	5
p1	1	0	1	2	3	4
p2	2	1	0	1	2	3
p3	3	2	1	0	1	2
p4	4	3	2	1	0	1
p5	5	4	3	2	1	0

Every step below shows this same information **indexed by logical qubits**: `D_layout[qa,qb] = D_phys[layout(qa),layout(qb)]`. Rows and columns stay in the order q1, q2, …, q6; the values change when their positions change. For example, initially `D_layout[q2,q3] = D_phys[p1,p3] = 2`. After SWAP(p2,p3), it becomes `D_phys[p1,p2] = 1`. The chip did not change; q3 moved. A gate emission changes neither matrix.

With those occupants:

Gate	Positions now	Distance	Adjacent?
g1 = CX(q2, q3)	p1, p3	2	No
g2 = CX(q5, q6)	p5, p2	3	No
g3 = CX(q1, q2)	p0, p1	1	Yes
g4 = CX(q3, q4)	p3, p4	1	Yes
g5 = CX(q2, q3)	p1, p3	2	No
g6 = CX(q4, q6)	p4, p2	2	No
g7 = CX(q5, q4)	p5, p4	1	Yes
g8 = CX(q4, q1)	p4, p0	4	No

g3, g4 and g7 are adjacent, but none of them can run yet: each is waiting for g1 or g2. That distinction is the subject of the next section.

5. Ready versus adjacent: the front layer and the lookahead layer

A two-qubit gate can be emitted only when **both** of these hold:

1. **Ready:** every gate it depends on has already been emitted.
2. **Adjacent:** its two qubits currently sit on connected positions.

Readiness comes from the program; adjacency comes from the layout. SWAPs change adjacency. Nothing changes readiness except emitting the predecessors. (“Emitting” a gate means writing it into the output circuit on the physical positions its qubits occupy at that moment. In circuit drawings a physical position is a horizontal **wire**, so “wire” and “position” are used interchangeably below.)

How readiness is tracked

Give each gate a counter: “how many of my predecessors are still not done”. The columns below follow the order in which the gates are actually emitted in sections 6 and 7:

Gate	At start	After g1	After g3	After g2, g4	After g6, g5	After g7
g1	0	done	done	done	done	done
g2	0	0	0	done	done	done
g3	1	0 → ready	done	done	done	done
g4	1	0 → ready	0	done	done	done
g5	2	2	1	0 → ready	done	done
g6	2	2	2	0 → ready	done	done
g7	2	2	2	1	0 → ready	done
g8	2	2	1	1	1	0 → ready

A gate with counter 0 is ready. Emitting a gate decrements the counters of its successors. (The code counts dependency *edges*, and two gates can be joined by more than one edge when they share two wires, but the idea is the same.)

The front layer

The **front layer**, written `F`, is the set of gates that are ready but **not** adjacent. These are the gates that routing must do something about.

Gates that are ready *and* adjacent never sit in `F`: the router emits them immediately, then checks whether that made anything else ready, and keeps going. Only when every ready gate is blocked does it stop and start choosing SWAPs. This "emit everything you can" sweep is why a circuit with thousands of gates may need only a few hundred SWAP decisions.

At the start of our example: g1 and g2 are ready, and neither is adjacent. `F = {g1, g2}`. This is the front layer the figure highlights. (The paper defines `F` as *all* ready gates; Qiskit's `F` holds only the ready gates that are blocked, because the others are emitted on the spot. At the start of this example the two definitions agree.)

The lookahead layer

The **lookahead layer**, written `E`, answers: "if the whole front layer were done, which two-qubit gates would become ready?" To compute it, the router copies the readiness counters, pretends every front gate is done, and collects the gates whose copied counter reaches 0. Nothing is actually emitted; the real counters are untouched.

At the start: pretending g1 and g2 are done makes g3 and g4 ready. `E = {g3, g4}`. g5 is not in `E`, because it also waits for g3 and g4, which are not done even in the pretend world; the same goes for g6, g7 and g8.

Three things about `E` are worth noticing:

- It can contain gates that are **already adjacent** (g3 and g4 both are). That is useful: a SWAP that pulled them apart would create work later, and the score should see that.
- The named `lookahead` and `decay` modes in this checkout use **one** lookahead layer. The Rust code supports more (each with its own weight), but the Python passes that call it do not use them.
- The code maintains `E` incrementally rather than recomputing it, and the incremental update has one gap: a gate whose last missing predecessor is emitted *during* an emit sweep, while another of its predecessors is sitting in `F`, is not added to `E`. Section 6.5 shows the one place this happens in our example (g5 after the first decision). It changes no decision there, but if you recompute `E` by hand from the definition you will see one more gate than the code does.

Reading a state snapshot

Read each figure from top to bottom: **layout, dependency graph, layers, distance matrix**. The graph keeps every gate in the same place across the walkthrough, so it is easy to see what changed. One-qubit gates remain omitted from this CX dependency graph.

Mark	Meaning
`F` in a node, dashed line underneath	Ready but blocked; the gates whose distances drive the next SWAP.
`E` in a node, *** underneath	Lookahead; scored but not permitted to run yet.
`done`, grey node	Already emitted in this pass; kept visible to preserve the dependency picture.
`wait`, plain node	Still waiting for predecessors, outside the current F and E.
`emit next`, green node	Ready and adjacent inside an emission sweep. No new SWAP is needed for it.
Green hardware edge and occupants	The SWAP just applied; green graph nodes instead mean executable gates.
Blue / amber matrix cells	The logical pairs currently scored in F / E.

Timing of the snapshots. State S0 is before decision 1. State S1 is after decision 1 **and its full emit sweep**, hence before decision 2; the same convention continues through S6. Separate intermediate snapshots inside decision 1 show

why “after the SWAP” and “after the emit sweep” are different moments. In an intermediate snapshot, a previously blocked F gate can be adjacent and awaiting immediate emission; no candidate is scored in that temporary state.

How to read a dependency edge. `g1 -> g3` labelled `q2,p1` says that g3 waits for g1 on q2, and q2 is **currently** at p1. It is not a device connection, and is not a record of where g1 ran in the past. Labels beside grey nodes also use the current layout. Use the emission tables for the physical operands at the time a gate was emitted.

The state before the first decision is S0. H(q1), Z(q1), H(q2), and H(q5) have already been emitted; no CX has been emitted yet.

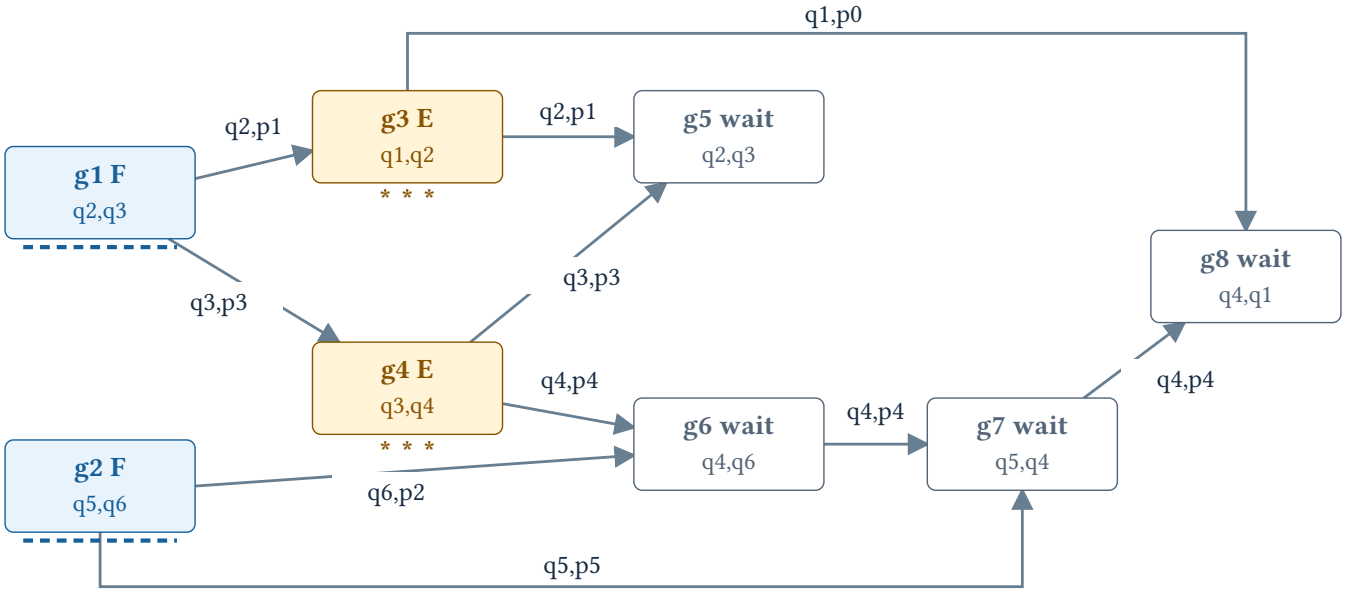
Routing state S0

Before decision 1. Initial one-qubit gates have already been emitted.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front ***** E: lookahead** done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	3	4	5	2
q2	1	0	2	3	4	1
q3	3	2	0	1	2	1
q4	4	3	1	0	1	2
q5	5	4	2	1	0	3
q6	2	1	1	2	3	0

Layers at this snapshot

F = {g1, g2}

E = {g3, g4}

Gate / physical operands / distance

g1: (p1,p3) d = 2

g2: (p5,p2) d = 3

g3: (p0,p1) d = 1

g4: (p3,p4) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 7: State S0, before decision 1: F={g1,g2}, E={g3,g4}. Every dependency edge includes a current physical location, and the matrix highlights the four scored pairs.

A quick reading check. g3 has distance 1, but its predecessor g1 is not grey: g3 cannot run. g1 has no unfinished predecessor, but distance 2: it needs routing. These are two different reasons for a gate to wait.

6. The first decision, step by step

6.1 Which SWAPs are candidates

Rule. A candidate is any hardware edge that has at least one endpoint occupied by a qubit of a front-layer gate. An edge touching two such qubits is listed once.

Why this rule. A SWAP that touches no front qubit cannot change any front distance in this step, so it cannot help now. Restricting candidates this way keeps every decision cheap: there are at most (2 × number of blocked gates × the largest number of neighbors any position has) candidates, however many edges the device has.

Example. The front gates use q2 (at p1), q3 (at p3), q5 (at p5) and q6 (at p2):

Front position	Its neighbors	Candidate edges
p1, holding q2	p0, p2	(p0, p1), (p1, p2)
p3, holding q3	p2, p4	(p2, p3), (p3, p4)
p5, holding q5	p4	(p4, p5)
p2, holding q6	p1, p3	(p1, p2), (p2, p3): already listed

Five candidates: `(p0, p1)`, `(p1, p2)`, `(p2, p3)`, `(p3, p4)`, `(p4, p5)`. On this small device that happens to be every edge; on a larger one, edges far from the front qubits would be left out.

Not a candidate: `(p1, p3)`, because it is not a hardware edge. A SWAP can only be applied to connected positions.

6.2 Score by front-layer distance

Rule. For each candidate, imagine applying it to the *current* layout (the candidates are alternatives, not a sequence), and compute

```
ΔD_F = (sum of front distances after the SWAP) − (sum of front distances now)
```

Smaller is better. A negative value is an improvement.

Example. The front has two gates, so each candidate is judged on the sum of both distances:

Candidate	Imagined occupants [p0 … p5]	d(q2, q3)	d(q5, q6)	ΔD_F
none (now)	[q1, q2, q6, q3, q4, q5]	2	3	—
SWAP(p0, p1)	[q2, q1, q6, q3, q4, q5]	3	3	+1
SWAP(p1, p2)	[q1, q6, q2, q3, q4, q5]	1	4	(1 − 2) + (4 − 3) = 0
SWAP(p2, p3)	[q1, q2, q3, q6, q4, q5]	1	2	(1 − 2) + (2 − 3) = **−2**
SWAP(p3, p4)	[q1, q2, q6, q4, q3, q5]	3	3	+1
SWAP(p4, p5)	[q1, q2, q6, q3, q5, q4]	2	2	−1

SWAP(p2, p3) is the clear winner: it moves q3 next to q2 *and* moves q6 one step toward q5. One SWAP, two gates helped. Compare SWAP(p1, p2), which also makes g1 adjacent but pushes q6 one step further from q5, for a net change of zero. Scoring the whole front rather than one gate at a time is what lets SABRE find moves like SWAP(p2, p3).

The code (`Layer::score`) does not recompute every distance. Only the two swapped positions move, so only their partners' distances can change; it adds up just those changes. That is the "delta" the name `ΔD_F` refers to.

6.3 Add the lookahead

Rule. Compute the same change for the lookahead layer, `ΔD_E`, and add a small multiple of it:

```
score = ΔD_F + w × ΔD_E,        w = 0.5 / (number of physical qubits)
```

The weight is small on purpose: the front layer must win when they disagree, and lookahead should only separate options that are equally good for the front. On our six-position device `w = 0.5 / 6 = 1/12 ≈ 0.0833`.

Example. The lookahead gates are g3 = CX(q1, q2) and g4 = CX(q3, q4). Now their distances sum to `1 + 1 = 2`.

Candidate	Imagined occupants	d(q1, q2)	d(q3, q4)	ΔD_E	ΔD_F	score = ΔD_F + ΔD_E / 12
SWAP(p0, p1)	[q2, q1, q6, q3, q4, q5]	1	1	0	+1	+1.0000
SWAP(p1, p2)	[q1, q6, q2, q3, q4, q5]	2	1	+1	0	+0.0833
SWAP(p2, p3)	[q1, q2, q3, q6, q4, q5]	1	2	+1	−2	**−1.9167**
SWAP(p3, p4)	[q1, q2, q6, q4, q3, q5]	1	1	0	+1	+1.0000
SWAP(p4, p5)	[q1, q2, q6, q3, q5, q4]	1	2	+1	−1	−0.9167

Winner: SWAP(p2, p3). The lookahead does not change the ranking here; it only records that the winner has a cost, namely moving q3 one step away from q4, which g4 will need later. A front gain of two steps outweighs a lookahead loss of one twelfth. Part III's extra terms are all zero at this point (nothing has been emitted, every wire is idle), so classic SABRE and this checkout make the same choice.

This is what lookahead is for: choose among equally good immediate moves using a glimpse of what comes next. It does not plan the whole future and it does not guarantee the globally best route; it is a cheap, informed preference.

6.4 Apply the winner

Only the winning SWAP is applied. The other candidates were hypothetical.

```
Insert SWAP(p2, p3) into the output.

Positions:   p0 ---- p1 ---- p2 ---- p3 ---- p4 ---- p5
Before:       q1      q2      q6      q3      q4      q5
After:        q1      q2      q3      q6      q4      q5
```

What changed and what did not:

Thing	Changed?
Hardware graph, distance table	No, never.
Layout (who is where)	Yes: q3 ⟶ p2, q6 ⟶ p3.
The gates still to route	No, but their physical positions must be re-read from the new layout.
Readiness counters	No. A SWAP emits no program gate.

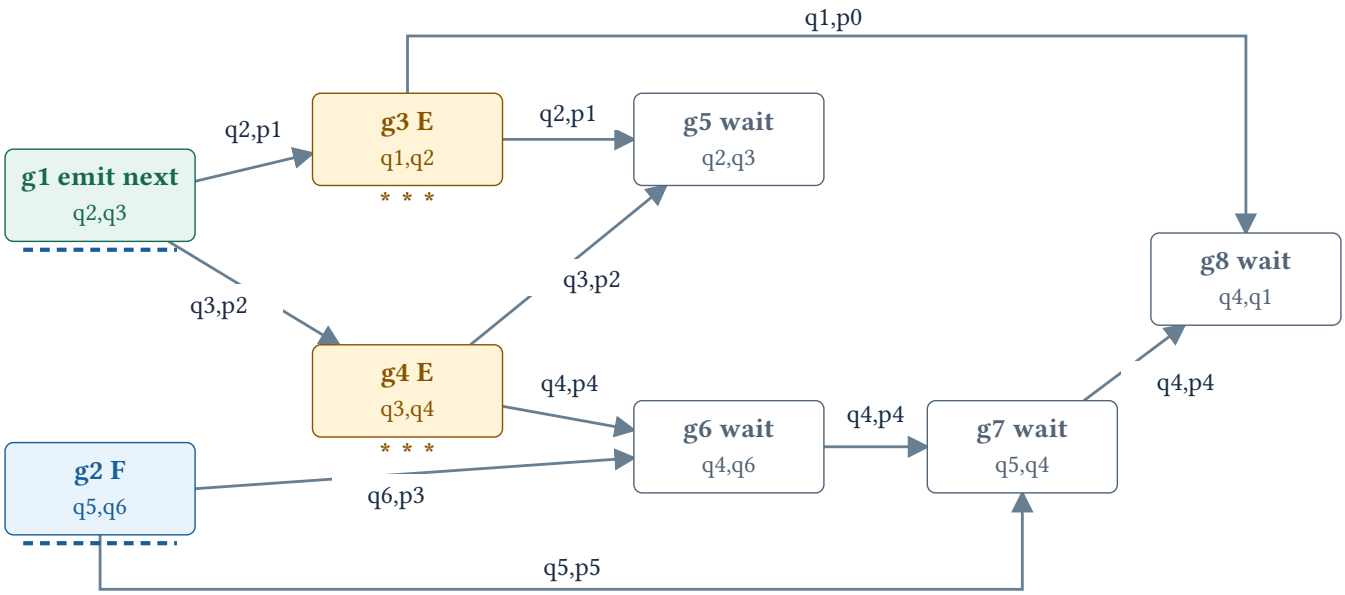
Decision 1 / immediately after the SWAP

SWAP(p2,p3) changed locations only. g1 is now executable; no CX has been emitted yet.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	2	4	5	3
q2	1	0	1	3	4	2
q3	2	1	0	2	3	1
q4	4	3	2	0	1	1
q5	5	4	3	1	0	2
q6	3	2	1	1	2	0

Layers at this snapshot

F = {g1, g2}

E = {g3, g4}

Gate / physical operands / distance

g1: (p1,p2) d = 1

g2: (p5,p3) d = 2

g3: (p0,p1) d = 1

g4: (p2,p4) d = 2

Edge labels give current locations, even beside completed gates. They are not hardware links.

Inside the emit sweep: the marked F/E are not yet a new scoring state. Green means executable.

Figure 8: Decision 1 immediately after SWAP(p2,p3): q3 is now at p2 and q6 at p3. g1 is executable, while the dependency graph has no completed CX yet.

Read the score directly from the two matrices. In S0 the front pairs had distances `D[q2,q3] + D[q5,q6] = 2 + 3 = 5`. In this intermediate snapshot they have `1 + 2 = 3`, so `ΔD_F = 3 - 5 = -2`. The lookahead pairs change from `1 + 1 = 2` to `1 + 2 = 3`, so `ΔD_E = +1`. Use the **same F and E before and after a hypothetical SWAP** for this calculation; do not substitute the new layers produced by emitting gates.

6.5 Emit what is now possible

g1 = CX(q2, q3): q2 at p1, q3 at p2, adjacent. Emit it as `CX(p1, p2)`. Its counter-decrement makes g3 and g4 ready:

Newly ready gate	Positions under [q1, q2, q3, q6, q4, q5]	Adjacent?	Action
g3 = CX(q1, q2)	p0, p1	Yes	Emit as CX(p0, p1).
g4 = CX(q3, q4)	p2, p4	No, distance 2	Put in the front layer.
g5 = CX(q2, q3), ready after g3 and g4	—	—	Still waits for g4.

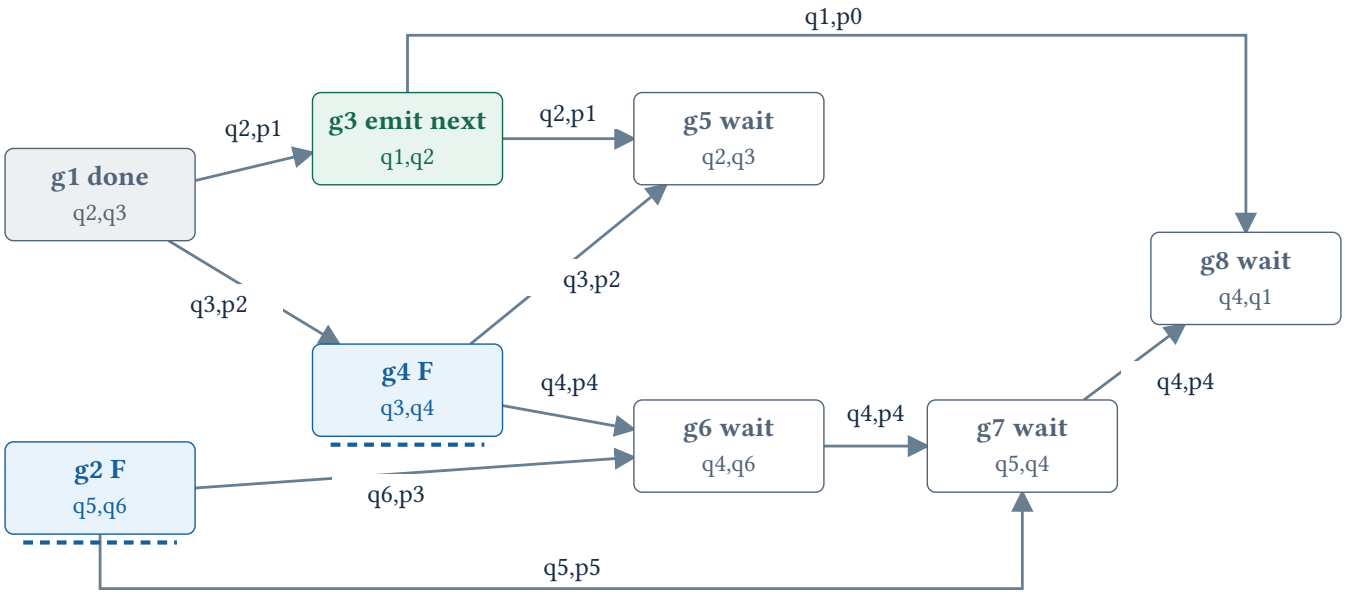
Decision 1 / after emitting g1

g3 is ready and adjacent: emit it next. Lookahead is rebuilt after the sweep.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	2	4	5	3
q2	1	0	1	3	4	2
q3	2	1	0	2	3	1
q4	4	3	2	0	1	1
q5	5	4	3	1	0	2
q6	3	2	1	1	2	0

Layers at this snapshot

F = {g2, g4}

E: rebuild pending

Gate / physical operands / distance

g2: (p5,p3) d = 2
g4: (p2,p4) d = 2
g3: (p0,p1) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

Inside the emit sweep: the marked F/E are not yet a new scoring state. Green means executable.

Figure 9: Inside decision 1, after emitting g1: g3 is executable, g4 joins the blocked front with g2, and the layout and distance matrix have not changed. Lookahead is rebuilt after the sweep.

g2 is still blocked (q5 at p5, q6 at p3, distance 2), so the front layer now holds two gates again: `F = {g2, g4}`.

The lookahead layer is rebuilt. Pretending g2 and g4 are done makes g6 ready (it waited for g4 and g2), so `E = {g6}`. By the definition, g5 should be there too: it waits for g3, which has just been emitted, and g4, which is in `F`. This is the gap mentioned in section 5. The code added g5′s last missing predecessor to the “done” side during the sweep that emitted g3, and the incremental update only adds a gate to `E` when the predecessor that completed its count is one that *stayed* in `F`. So

the code's `E` is `{g6}`. For the next decision this makes no difference: with g5 included, SWAP(p2, p3) would score one twelfth worse and nothing else would change, and SWAP(p2, p3) does not win anyway.
State S1 is after the first decision and the complete emission sweep:

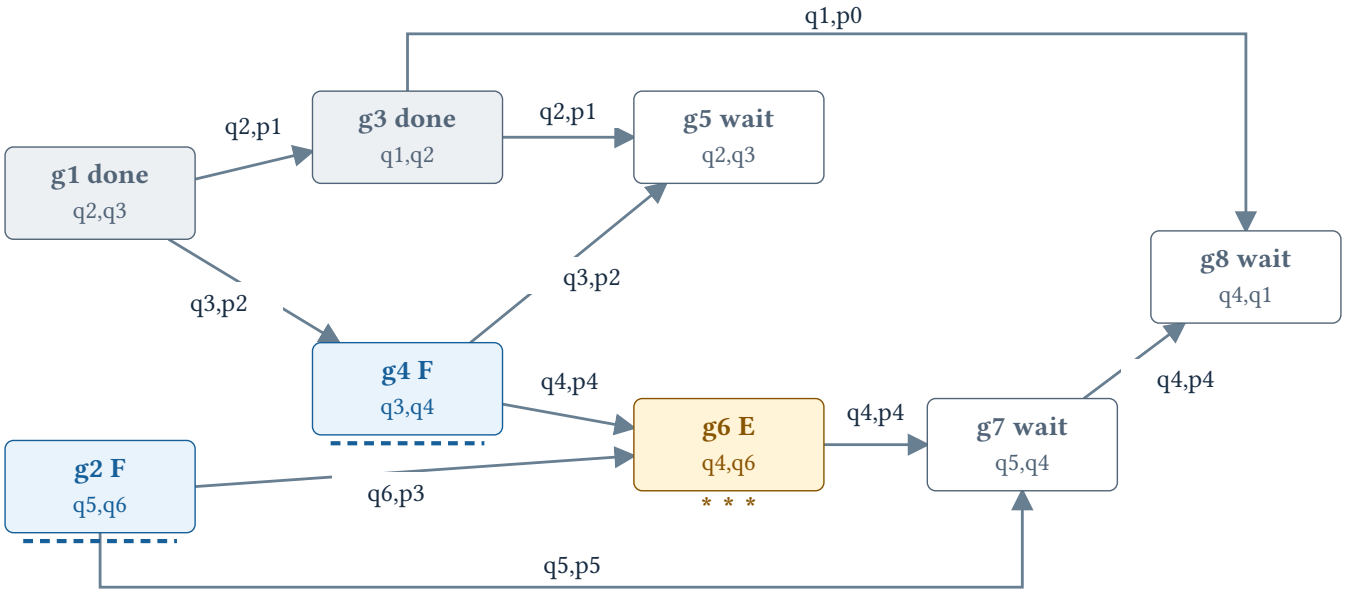
Routing state S1

After decision 1: emit g1, then g3. Next: decision 2.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	2	4	5	3
q2	1	0	1	3	4	2
q3	2	1	0	2	3	1
q4	4	3	2	0	1	1
q5	5	4	3	1	0	2
q6	3	2	1	1	2	0

Layers at this snapshot

F = {g2, g4}
E = {g6}

Gate / physical operands / distance

g2: (p5,p3) d = 2
g4: (p2,p4) d = 2
g6: (p4,p3) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.
E matches this checkout: g5 is omitted by its incremental update (section 6.5).

Figure 10: State S1, before decision 2: g1 and g3 are done; F={g2,g4}, E={g6}. g5 stays unstarred to match this checkout's incremental lookahead. The matrix is the same as immediately after the SWAP.

Notice that the front layer did not simply inherit the old lookahead layer. g3 was adjacent, so it was emitted straight away; g4 was not, so it joined g2 in the front. The lookahead layer was then rebuilt from the new front.

7. Finishing the circuit: decisions 2 to 6

Decision 2

Starting snapshot: S1 above.

Front: g2 with q5 at p5 and q6 at p3; g4 with q3 at p2 and q4 at p4. Every position from p2 to p5 holds a front qubit, so the candidates are the edges touching any of them: `(p1, p2)`, `(p2, p3)`, `(p3, p4)` and `(p4, p5)`. The edge `(p0, p1)` touches no front qubit and is not considered. Lookahead: g6 = CX(q4, q6), with q4 at p4 and q6 at p3, distance 1.

Candidate	Imagined occupants	d(q5, q6)	d(q3, q4)	ΔD_F	d(q4, q6)	ΔD_E	score
SWAP(p1, p2)	[q1, q3, q2, q6, q4, q5]	2	3	+1	1	0	+1.0000
SWAP(p2, p3)	[q1, q2, q6, q3, q4, q5]	3	1	0	2	+1	+0.0833
SWAP(p3, p4)	[q1, q2, q3, q4, q6, q5]	1	1	**−2**	1	0	**−2.0000**
SWAP(p4, p5)	[q1, q2, q3, q6, q5, q4]	1	3	0	2	+1	+0.0833

Winner: SWAP(p3, p4), and again one SWAP serves both front gates: it carries q6 to p4, next to q5, and q4 to p3, next to q3. (This checkout also adds ready-time penalties of at most 0.0001 here, section 12; they do not change the order.)

After the SWAP the occupants are `[q1, q2, q3, q4, q6, q5]`, and the emit sweep is the longest of the whole run:

Gate	Positions	Adjacent?	Action
g2 = CX(q5, q6)	p5, p4	Yes	Emit as CX(p5, p4).
g4 = CX(q3, q4)	p2, p3	Yes	Emit as CX(p2, p3), followed by H on p2 and H on p3 (the one-qubit gates that come after g4 on q3 and q4).
g6 = CX(q4, q6), ready after g4 and g2	p3, p4	Yes	Emit as CX(p3, p4), followed by Z on p3.
g5 = CX(q2, q3), ready after g3 and g4	p1, p2	Yes	Emit as CX(p1, p2).
g7 = CX(q5, q4), ready after g2 and g6	p5, p3	No, distance 2	Put in the front layer.
g8 = CX(q4, q1), ready after g7 and g3	—	—	Still waits for g7.

Four CX gates and three one-qubit gates come out after a single SWAP. This is the “emit everything you can” sweep of section 5 doing its job.

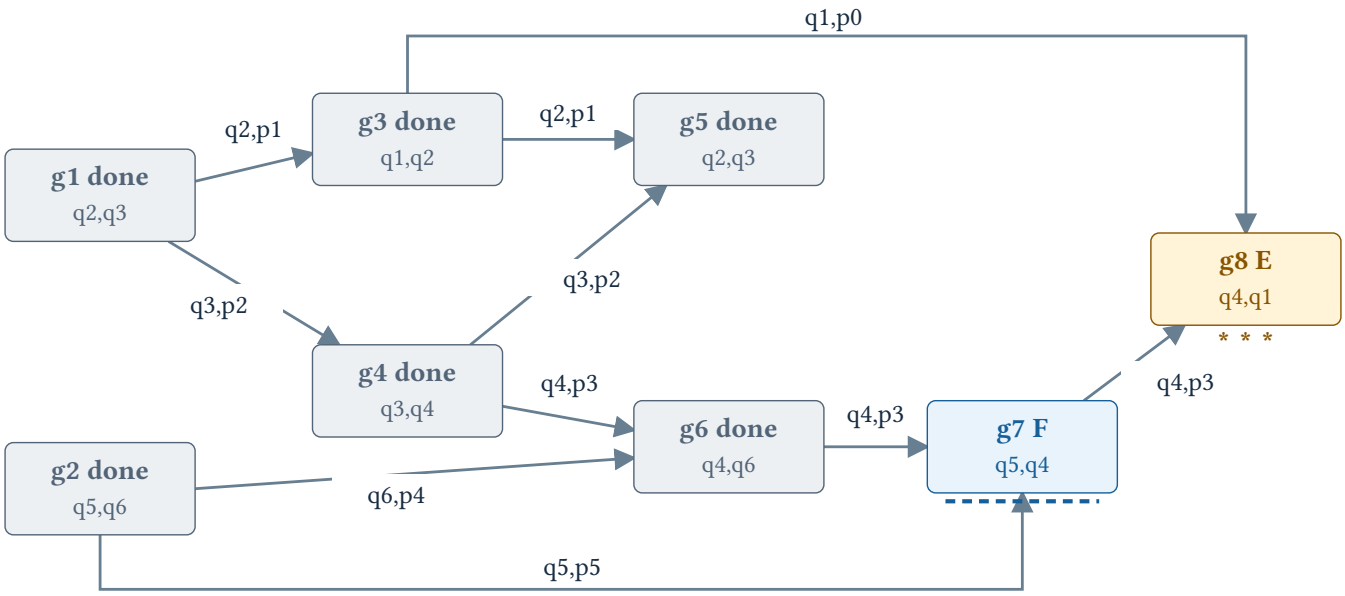
Routing state S2

After decision 2: emit g2, g4, g6 and g5. Next: decision 3.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	2	3	5	4
q2	1	0	1	2	4	3
q3	2	1	0	1	3	2
q4	3	2	1	0	2	1
q5	5	4	3	2	0	1
q6	4	3	2	1	1	0

Layers at this snapshot

F = {g7}

E = {g8}

Gate / physical operands / distance

g7: (p5,p3) d = 2

g8: (p3,p0) d = 3

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 11: State S2, before decision 3: g1 through g6 are done; F={g7}, E={g8}. The matrix shows d(q5,q4)=2 and d(q4,q1)=3.

Decision 3

Starting snapshot: S2 above.

Front: g7 with q5 at p5 and q4 at p3. Candidates: edges touching p5 or p3, which are `(p4, p5)`, `(p2, p3)` and `(p3, p4)`. Lookahead: g8, with q4 at p3 and q1 at p0, distance 3.

Candidate	Imagined occupants	d(q5, q4)	ΔD_F	d(q4, q1)	ΔD_E	score (classic SABRE)
SWAP(p2, p3)	[q1, q2, q4, q3, q6, q5]	3	+1	2	−1	+0.9167
SWAP(p3, p4)	[q1, q2, q3, q6, q4, q5]	1	−1	4	+1	−0.9167
SWAP(p4, p5)	[q1, q2, q3, q4, q5, q6]	1	−1	3	0	**−1.0000**

Two candidates make g7 adjacent. Classic SABRE prefers SWAP(p4, p5), which moves q5 toward q4 and leaves q4 where it is, over SWAP(p3, p4), which moves q4 one step further from q1 and so costs one twelfth on the lookahead.

This checkout chooses SWAP(p3, p4) instead. The gate emitted immediately before on that same pair of wires was g6 = CX(p3, p4), and a SWAP that directly follows a CX on the same pair is cheaper than a SWAP elsewhere: a later optimization pass merges the two into two CX gates instead of four. The *fusion credit* of section 11 gives SWAP(p3, p4) a bonus of −2/3, which is much more than the one twelfth it loses on the lookahead. The ready-time penalties of section 12 are equal for the two and do not matter. Take the numbers on trust for now; section 11 works them out. This is the one decision in the run where the extra terms do more than break a tie: they overturn the classic ranking. We follow it.

After SWAP(p3, p4) the occupants are `[q1, q2, q3, q6, q4, q5]`. Yes, this undoes the previous SWAP(p3, p4) as far as the layout is concerned, but the computation is **not** undone: g2, g4, g6 and g5 were all emitted between the two SWAPs, and g6 used exactly that pair of wires. g7 = CX(q5, q4) is now on p5 and p4, adjacent, and is emitted as `CX(p5, p4)`. That makes g8 ready: q4 at p4 and q1 at p0, distance 4, so it goes into the front layer. Nothing comes after g8, so `E` is empty.

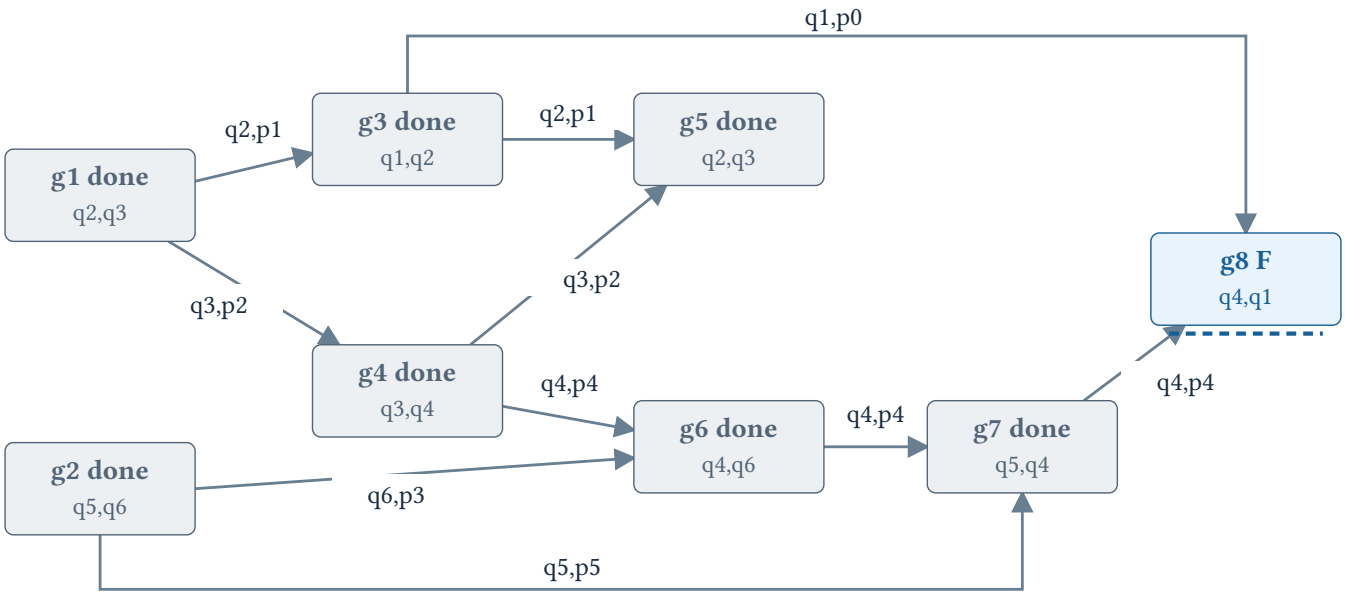
Routing state S3

After decision 3: emit g7. Next: decision 4.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	2	4	5	3
q2	1	0	1	3	4	2
q3	2	1	0	2	3	1
q4	4	3	2	0	1	1
q5	5	4	3	1	0	2
q6	3	2	1	1	2	0

Layers at this snapshot

F = {g8}

E = {}

Gate / physical operands / distance

g8: (p4,p0) d = 4

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 12: State S3, before decision 4: g7 is now done, g8 is the only front gate, and E is empty. The matrix shows d(q4,q1)=4.

Same layout, different progress. S3 has the same occupants and matrix as S1, but five more gates are grey. A layout alone is not the routing state: you also need to know which gates are done and which remain in F and E.

Decisions 4, 5 and 6

g8 needs three more SWAPs: q4 and q1 are four steps apart, and a SWAP moves one of them one step. With `E` empty the lookahead term is zero throughout, and at every one of the three decisions two candidates tie on distance. In classic SABRE each would be a coin flip. This checkout breaks all three with the ready-time penalty of section 12, which prefers the SWAP whose wires have been idle longest; section 12 shows the numbers. Here is the distance picture.

Decision 4. q4 at p4, q1 at p0. Candidates: `(p3, p4)`, `(p4, p5)` and `(p0, p1)`.

Candidate	Moves	d(q4, q1)	ΔD_F	Decided by
SWAP(p0, p1)	q1 → p1	3	−1	**wins**: p0 and p1 have been idle since g3
SWAP(p3, p4)	q4 → p3	3	−1	loses: p3 and p4 were just used by the last SWAP and g7
SWAP(p4, p5)	q4 → p5	5	+1	never in contention (its fusion credit after g7 only softens +1 to +1/3)

Occupants become `[q2, q1, q3, q6, q4, q5]`. g8 is still blocked (distance 3), so no gate is emitted and the search continues from the same front.

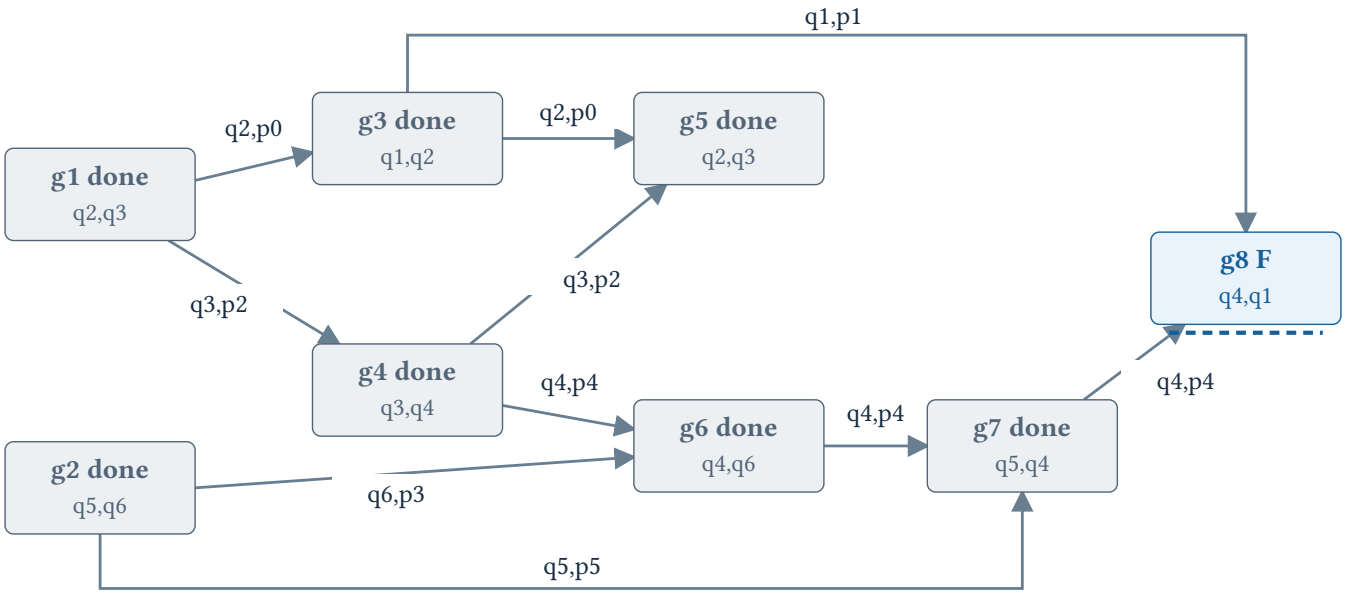
Routing state S4

After decision 4: no CX emitted; g8 is still blocked. Next: decision 5.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	1	3	4	2
q2	1	0	2	4	5	3
q3	1	2	0	2	3	1
q4	3	4	2	0	1	1
q5	4	5	3	1	0	2
q6	2	3	1	1	2	0

Layers at this snapshot

F = {g8}

E = {}

Gate / physical operands / distance

g8: (p4,p1) d = 3

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 13: State S4, before decision 5: SWAP(p0,p1) moves q1 to p1 and q2 to p0. No new gate is done; F={g8}, E is empty, and d(q4,q1)=3.

Decision 5. q4 at p4, q1 at p1. Candidates: `(p3, p4)`, `(p4, p5)`, `(p0, p1)` and `(p1, p2)`.

Candidate	Moves	d(q4, q1)	ΔD_F	Decided by
SWAP(p1, p2)	q1 ⟶ p2	2	−1	**wins**: p2 has been idle since g5
SWAP(p3, p4)	q4 ⟶ p3	2	−1	loses on ready time, by a hair
SWAP(p0, p1)	q1 back to p0	4	+1	
SWAP(p4, p5)	q4 ⟶ p5	4	+1 (+1/3 after credit)	

Occupants become `[q2, q3, q1, q6, q4, q5]`. Distance 2: still blocked.

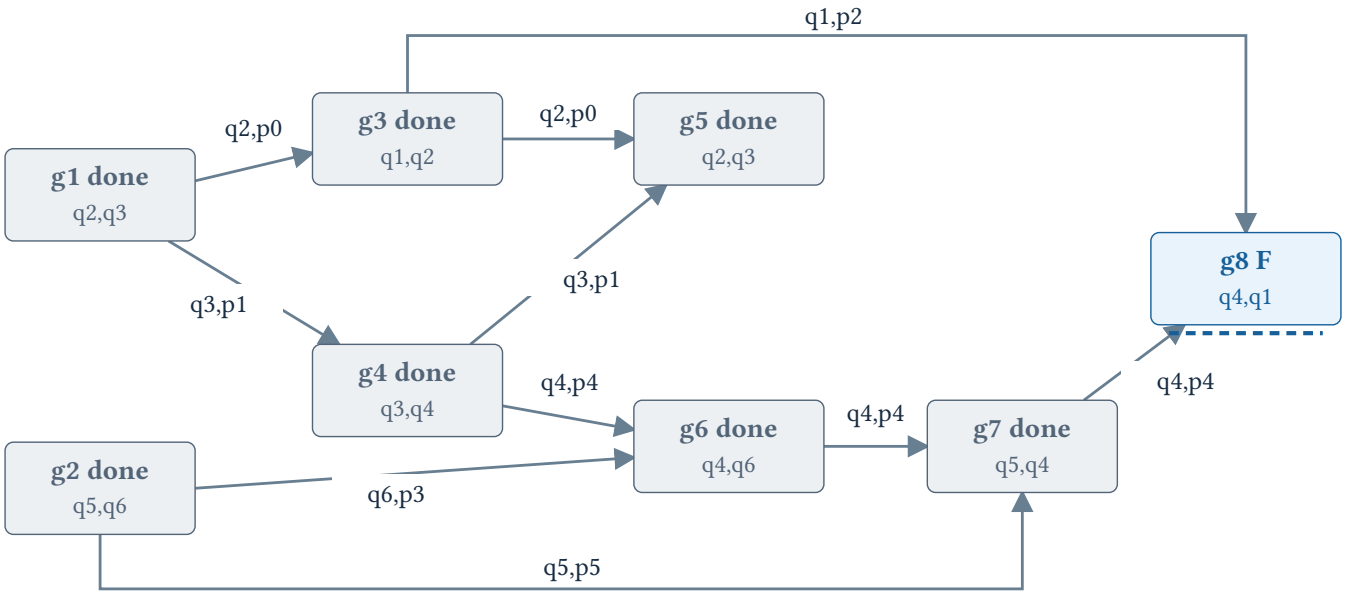
Routing state S5

After decision 5: no CX emitted; g8 is still blocked. Next: decision 6.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	2	1	2	3	1
q2	2	0	1	4	5	3
q3	1	1	0	3	4	2
q4	2	4	3	0	1	1
q5	3	5	4	1	0	2
q6	1	3	2	1	2	0

Layers at this snapshot

F = {g8}

E = {}

Gate / physical operands / distance

g8: (p4,p2) d = 2

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 14: State S5, before decision 6: SWAP(p1,p2) moves q1 to p2 and q3 to p1. F={g8}, E is empty, and d(q4,q1)=2.

Decision 6. q4 at p4, q1 at p2. Candidates: `(p3, p4)`, `(p4, p5)`, `(p1, p2)` and `(p2, p3)`.

Candidate	Moves	d(q4, q1)	ΔD_F	Decided by
SWAP(p3, p4)	q4 ⟶ p3	1	−1	**wins**: now p3, p4 are the fresher pair
SWAP(p2, p3)	q1 ⟶ p3	1	−1	loses: p2 was just used by SWAP(p1, p2)
SWAP(p1, p2)	q1 back to p1	3	+1	
SWAP(p4, p5)	q4 ⟶ p5	3	+1 (+1/3 after credit)	

Occupants become `[q2, q3, q1, q4, q6, q5]`. q4 at p3 and q1 at p2 are adjacent, and g8 is emitted as `CX(p3, p2)`. The front layer is empty, so routing is finished.

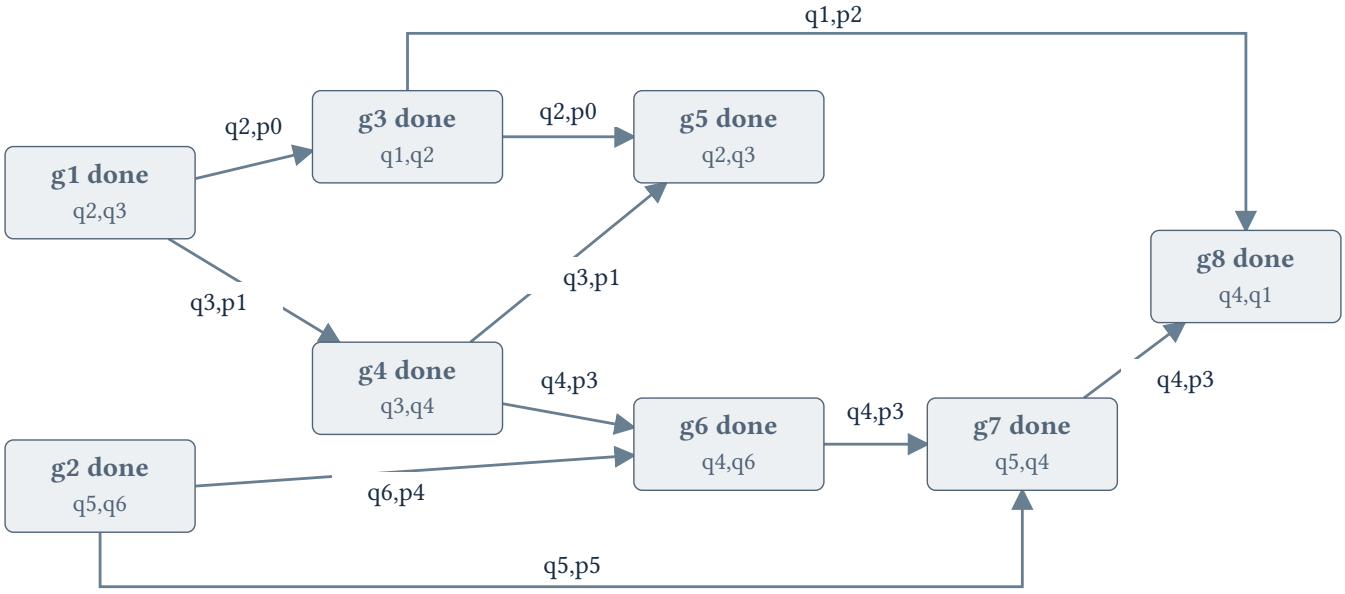
Routing state S6

After decision 6: emit g8. Routing is complete.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	2	1	1	3	2
q2	2	0	1	3	5	4
q3	1	1	0	2	4	3
q4	1	3	2	0	2	1
q5	3	5	4	2	0	1
q6	2	4	3	1	1	0

Layers at this snapshot

F = {}

E = {}

Gate / physical operands / distance

No gates remain to route.

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 15: State S6, after decision 6 and emission of g8: all eight CX gates are done. F and E are empty; the final matrix has d(q4,q1)=1.

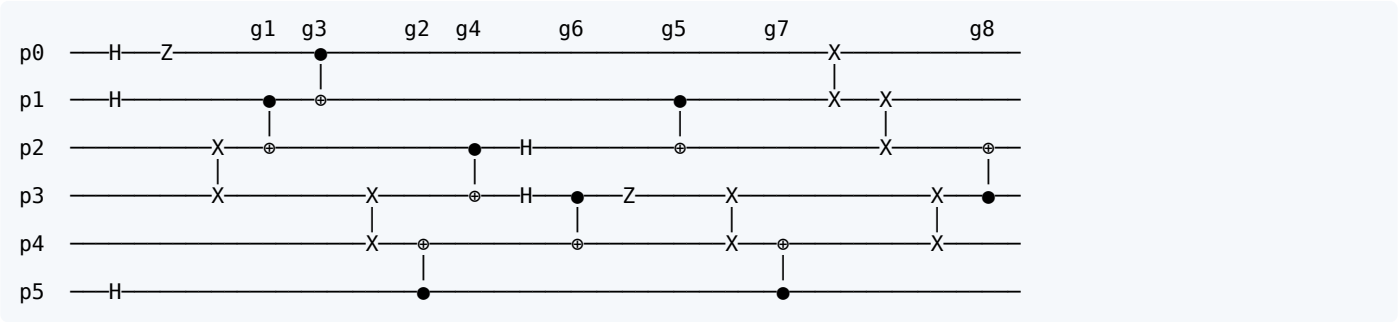
The pattern is worth a second look. q1 walked two steps to the right on wires that had been idle since early in the circuit, and q4 took one step to the left at the end. Each of those SWAPs can start as soon as its own wires are free, rather than queueing behind the busy wires p3–p5, which is exactly what the ready-time term is designed to encourage.

The complete route

The state sequence is `S0 -> S1 -> S2 -> S3 -> S4 -> S5 -> S6`. Each arrow is one chosen SWAP followed by every gate it enables. The two arrows into S4 and S5 emit no program gate.

Event	Occupants afterward	Emitted	Still blocked
Start	[q1, q2, q6, q3, q4, q5]	H(p0), Z(p0), H(p1), H(p5)	g1 at distance 2, g2 at distance 3
Decision 1	[q1, q2, q3, q6, q4, q5]	SWAP(p2, p3)	g2 at distance 2
Emit	[q1, q2, q3, q6, q4, q5]	CX(p1, p2) = g1, CX(p0, p1) = g3	g4 at distance 2 joins g2
Decision 2	[q1, q2, q3, q4, q6, q5]	SWAP(p3, p4)	nothing in front of g2 and g4
Emit	[q1, q2, q3, q4, q6, q5]	CX(p5, p4) = g2; CX(p2, p3) = g4, H(p2), H(p3); CX(p3, p4) = g6, Z(p3); CX(p1, p2) = g5	g7 at distance 2
Decision 3	[q1, q2, q3, q6, q4, q5]	SWAP(p3, p4)	nothing in front of g7
Emit	[q1, q2, q3, q6, q4, q5]	CX(p5, p4) = g7	g8 at distance 4
Decision 4	[q2, q1, q3, q6, q4, q5]	SWAP(p0, p1)	g8 at distance 3
Decision 5	[q2, q3, q1, q6, q4, q5]	SWAP(p1, p2)	g8 at distance 2
Decision 6	[q2, q3, q1, q4, q6, q5]	SWAP(p3, p4)	nothing
Emit	[q2, q3, q1, q4, q6, q5]	CX(p3, p2) = g8	finished

The physical output circuit (X–X = SWAP, ● = control, ⊕ = target):



Six SWAPs, eight CX gates. Expanding each SWAP into three CX gives 26 CX gates before any optimization; later passes can reduce that (section 11).

Final layout: `q1 → p2, q2 → p0, q3 → p1, q4 → p3, q5 → p5, q6 → p4`, that is, occupants `[q2, q3, q1, q4, q6, q5]`. The output does not swap everyone back to where they started. Qiskit records this final layout so that later passes and measurement results are interpreted correctly: a measurement of program qubit q1 must read position **p2**.

Why this route is legal (worth checking for any hand-worked example):

1. Every SWAP and every CX uses a hardware edge.
2. Every CX acts on the positions that hold its program qubits *at that moment*, with control and target in the right order.
3. No dependency is violated: g1 before g3 and g4; g3 and g4 before g5; g2 and g4 before g6; g2 and g6 before g7; g7 and g3 before g8. The one-qubit gates stay on their own qubit's wire and in their original order relative to the CX gates on that qubit.
4. The final layout accounts for every SWAP.

A route can be legal and still wasteful. Legality is a correctness property; SWAP count and depth are quality properties.

8. Decay: spread the SWAPs over different wires

The idea: when two candidate SWAPs make similar progress, prefer the one whose physical wires have not just been used by another SWAP. Distance and lookahead describe where the qubits are; **decay** adds a small memory of how they got there. A recently used wire makes a candidate slightly more expensive.

This section first explains decay with only the classic distance and lookahead terms. Section 8.6 then adds this checkout's fusion and ready-time terms to reproduce its exact choice. The increment `0.001`, reset interval `5`, and weight `w = 0.5 / number of physical qubits` below are this checkout's Python wrappers' decay configuration (section 13), not universal constants of SABRE.

8.1 What does “spread over different wires” mean?

A **wire** here means a fixed physical position, such as p1, not whichever program qubit currently occupies it. **SWAPs that share a physical qubit must run one after another; SWAPs on separate qubits can run simultaneously.** Decay encourages choices that allow this parallel execution.

Consider a small example with four physical positions and a gate needed between program qubits **A and B**:

```
Position:   p0 ---- p1 ---- p2 ---- p3
Occupant:    A       X       Y       B
```

Suppose SABRE first chooses `SWAP(p0, p1)`:

```
Position:   p0 ---- p1 ---- p2 ---- p3
Occupant:    X       A       Y       B
```

Now two choices bring A and B next to each other:

Next SWAP	Resulting occupants, p0 to p3	Relationship to the first SWAP
SWAP(p1, p2)	`[X, Y, A, B]`	Shares p1, so must wait for the first SWAP to finish.
SWAP(p2, p3)	`[X, A, B, Y]`	Uses separate wires, so can run alongside the first SWAP.

Both routes use **two SWAPs** and enable the desired gate. Assuming the wires were initially free and each SWAP takes one time slot, their execution can look like this:

```
Reuse p1:
    Time slot 1:  SWAP(p0, p1)
    Time slot 2:  SWAP(p1, p2)
    Then:         gate on A and B

Use separate wires:
    Time slot 1:  SWAP(p0, p1) AND SWAP(p2, p3)
    Then:         gate on A and B
```

SABRE choosing a SWAP second does not mean the device must execute it second. The compiler constructs the circuit sequentially, but independent operations can execute together. Reusing p1 extends a chain of operations that must run in sequence; choosing disjoint wires allows the same SWAP count with less **depth** in this example. The original SABRE paper introduces decay to encourage this trade-off between gate count and depth.

Decay does not build a schedule or know when a wire is actually free. It uses recent SWAP use as a cheap hint: in general, separate wires can overlap only if their earlier dependencies allow it. Fresh wires are only a preference among legal candidates, and reusing a wire can still win if it makes enough additional routing progress. Decay does not add hardware edges to the candidate list.

8.2 The memory: one multiplier per physical position

Call the stored multiplier at position p `decay[p]`. Initially every entry is `1.000`, which means “no extra cost”. After choosing and applying a search SWAP, the router normally **adds** `0.001` to each endpoint’s entry. For example, if no reset intervenes:

```
Physical position:          p0     p1     p2     p3     p4     p5
Before any SWAP:         1.000  1.000  1.000  1.000  1.000  1.000
After SWAP(p0, p1):      1.001  1.001  1.000  1.000  1.000  1.000
Then SWAP(p1, p2):       1.001  1.002  1.001  1.000  1.000  1.000
```

p1 was used twice, so its entry is `1 + 2 × 0.001 = 1.002`. This is an additive update, not repeated multiplication by `1.001`. The entries stay attached to their physical positions when program qubits trade places.

For a candidate `SWAP(a, b)`, use just **one** multiplier:

```
m = max(decay[a], decay[b])
```

`max` means “take the larger of the two numbers”. Immediately after the *first* SWAP in the table:

Candidate	Endpoint entries	m	Reason
SWAP(p1, p2)	1.001, 1.000	1.001	Reusing even one endpoint incurs the penalty.
SWAP(p3, p4)	1.000, 1.000	1.000	Neither endpoint has been used since the last reset.
SWAP(p0, p1)	1.001, 1.001	1.001	Two used endpoints do not double the penalty.

This maximum is a simple way to let the more heavily reused endpoint determine the penalty. It is neither the sum nor the product of the endpoint entries.

Scoring does not update the memory. All candidates in one decision read the same stored entries. Only after the winner is selected do its endpoints get updated. There are two reset rules, explained in section 8.7: routing a gate clears the memory, and every fifth search SWAP clears it instead of incrementing its endpoints. That is why the table above assumes no reset.

8.3 The formula: multiply the distance left after the candidate

Before adding decay, sections 6 and 7 compared **changes** in distance. For decay we need the **distance total after** each candidate instead. Keep F and E fixed while comparing candidates; do not emit a newly adjacent gate or rebuild the lookahead inside this calculation.

Symbol	Meaning at the current decision
`D_F`	Sum of the current distances of all front-layer pairs.
`D_E`	Sum of the current distances of all lookahead pairs; 0 if E is empty.
`w`	Lookahead weight; `1/12` on our six-position device.
`ΔD_F`	Front sum after the candidate minus front sum now.
`ΔD_E`	Lookahead sum after the candidate minus lookahead sum now.
`H_now`	Current weighted total, `D_F + w × D_E`. Same for every candidate.
`H_after`	Weighted total if this candidate were applied. Can differ by candidate.
`m`	Larger stored decay entry of this candidate's two endpoints.

Read the calculation as three steps:

```
1. Current total:      H_now   = D_F + w × D_E
2. Candidate's total: H_after = H_now + ΔD_F + w × ΔD_E
3. Penalize reuse:    score   = H_after × m
```

Step 2 is simply "current amount plus change equals new amount". Expanding it shows why it is the post-SWAP distance total:

```
H_after = (D_F + ΔD_F) + w × (D_E + ΔD_E)
        = front sum after + w × lookahead sum after
```

Here **absolute score** means this total, as opposed to a change. It does **not** mean applying an absolute-value operation such as `abs(ΔD_F)`. The score is also not a count of remaining SWAPs: adjacent pairs contribute distance 1, not 0, and one SWAP can affect several pairs.

Why not multiply the change directly? Suppose two candidates each improve the weighted distance by 1, so each has change `-1`. One uses a fresh pair (`m = 1`); the other reuses a wire (`m = 1.001`). Multiplying those changes would give `-1` and `-1.001`. Smaller wins, so this would *reward* reuse.

Instead, suppose `H_now = 3`. Both candidates leave `H_after = 3 - 1 = 2`:

```
Fresh pair:   (3 - 1) × 1.000 = 2.000
Reused wire:  (3 - 1) × 1.001 = 2.002
```

Now reuse raises the score, which is the intended penalty. The extra cost is `H_after × (m - 1)`: here `2 × 0.001 = 0.002`. A multiplier of `1.001` adds **0.1% of the post-SWAP total**, not a fixed `0.001` score penalty.

Without decay, adding the same `H_now` to every candidate cannot change their ranking. With decay, candidates can have different multipliers, so that baseline must be included **before** multiplying. In particular, `(H_now + change) × m` is not the same as `H_now + change × m`.

8.4 Worked example: decision 1, before any wire has been reused

At S0, the layout is `[q1, q2, q6, q3, q4, q5]`. The two front gates are g1 = CX(q2, q3) and g2 = CX(q5, q6); the lookahead contains g3 = CX(q1, q2) and g4 = CX(q3, q4). Therefore:

```
D_F   = d(q2, q3) + d(q5, q6) = 2 + 3 = 5
D_E   = d(q1, q2) + d(q3, q4) = 1 + 1 = 2
w     = 0.5 / 6 = 1/12
H_now = 5 + (1/12) × 2 = 31/6 = 5.166666...
```

Take the eventual winner, `SWAP(p2, p3)`, slowly. Its imagined layout is `[q1, q2, q3, q6, q4, q5]`. The front distances become 1 and 2, so `ΔD_F = (1 + 2) - 5 = -2`. The lookahead distances become 1 and 2, so `ΔD_E = (1 + 2) - 2 = +1`. Substituting:

```
H_after = 31/6 - 2 + (1/12) × 1 = 39/12 = 3.25
m       = max(decay[p2], decay[p3]) = max(1, 1) = 1
score   = 3.25 × 1 = 3.25
```

You can check the same result directly from the new distances: `H_after = (1 + 2) + (1/12) × (1 + 2) = 3.25`. g1 still contributes its distance of 1 during scoring; it is only emitted after the winner is applied.

Here are all five alternatives. The post-SWAP sums come from section 6, and the decimal scores are rounded to four places only for display:

Candidate	Front sum after	Lookahead sum after	H_after	m	Decay score
SWAP(p0, p1)	6	2	6 + 2/12	1	6.1667
SWAP(p1, p2)	5	3	5 + 3/12	1	5.2500
SWAP(p2, p3)	3	3	3 + 3/12	1	**3.2500**
SWAP(p3, p4)	6	2	6 + 2/12	1	6.1667
SWAP(p4, p5)	4	3	4 + 3/12	1	4.2500

Every multiplier is 1, so there is no history-based preference yet. The ranking is exactly the same as section 6's change-based ranking. This checkout's fusion credit and ready penalty are also zero at this decision, so `SWAP(p2, p3)` wins with the complete score too.

8.5 Worked example: decision 5, where decay changes the choice

Follow this checkout's forward route with decay enabled, as `SabreLayout` does in section 16. It reaches the same S4 as section 7: decisions 1 through 3 emitted gates, but decision 4 inserted `SWAP(p0, p1)` and emitted none. The decay memory from decision 4 therefore survives into decision 5:

```
Position:     p0 ---- p1 ---- p2 ---- p3 ---- p4 ---- p5
Occupant:     q2      q1      q3      q6      q4      q5
Decay:       1.001   1.001   1.000   1.000   1.000   1.000

F = {g8 = CX(q4, q1)}, E = empty
D_F = |4 - 1| = 3, D_E = 0, H_now = 3
```

The important comparison is between moving q1 right and moving q4 left. Both shorten the distance from 3 to 2; neither makes g8 executable yet.

Candidate	Movement	Distance after	H_after	m	Classic decay score
SWAP(p1, p2)	q1: p1 to p2	2	3 - 1 = 2	1.001	2.002
SWAP(p3, p4)	q4: p4 to p3	2	3 - 1 = 2	1.000	**2.000**
SWAP(p0, p1)	q1: p1 back to p0	4	3 + 1 = 4	1.001	4.004
SWAP(p4, p5)	q4: p4 to p5	4	3 + 1 = 4	1.000	4.000

For the first candidate, p1 was used by decision 4: `m = max(1.001, 1.000) = 1.001`, so the score is `2 × 1.001 = 2.002`. For the second, neither p3 nor p4 has been used by a SWAP **since the last decay reset**: `m = max(1, 1) = 1`, so the score is `2 × 1 = 2`. p3 and p4 did participate in earlier SWAPs, but emitting gates cleared that history. A value of 1 does not mean "never used in the whole circuit".

Decay prefers SWAP(p3, p4). It makes the same distance progress without reusing p1. The two other candidates move the pair apart and are much worse. No randomness is needed to settle this comparison.

8.6 Adding this checkout's extra terms to decision 5

The previous table isolates decay. To reproduce the exact numbers from this checkout, include the two terms explained fully in sections 11 and 12:

- **Fusion credit `f`:** a negative adjustment when a SWAP can combine with a preceding two-qubit block on the same pair. Here only `(p4, p5)` gets a credit, `-2/3`, because g7 was just emitted on that pair.
- **Ready penalty:** `0.000022 × T`, where `T` is the larger of the two endpoints' estimated ready times. It slightly favors a pair that can start sooner. Here `ready = [11, 11, 8, 11, 12, 12]`, in p0 through p5 order.

In this decay configuration the front weight is 1, so the full formula is:

```
score = (H_now + ΔD_F + w × ΔD_E + f) × m + 0.000022 × T
      = (H_after + f) × m                 + ready penalty
```

The fusion credit is **inside** the multiplication. The ready penalty is added **after** it. At decision 5 the lookahead is empty, so its term is zero.

Candidate	H_after + f	m	After multiplication	Ready penalty	Total
SWAP(p1, p2)	2 + 0 = 2	1.001	2.002	0.000242	2.002242
SWAP(p3, p4)	2 + 0 = 2	1.000	2.000	0.000264	**2.000264**
SWAP(p0, p1)	4 + 0 = 4	1.001	4.004	0.000242	4.004242
SWAP(p4, p5)	4 - 2/3 = 10/3	1.000	3.333333…	0.000264	3.333597

For `(p1, p2)`, `T = max(11, 8) = 11`, giving `0.000022 × 11 = 0.000242`. For `(p3, p4)`, `T = max(11, 12) = 12`, giving `0.000264`. Its ready penalty is larger by just `0.000022`, whereas avoiding p1 saves `0.002` in decay cost. Thus `(p3, p4)` still wins, by `2.002242 - 2.000264 = 0.001978`.

This explains the difference from Part II, which used lookahead mode without decay. There, the two leading candidates had scores `-1 + 0.000242` and `-1 + 0.000264`, so the smaller ready penalty picked `(p1, p2)`. With decay enabled, the preference for avoiding a recently used wire outweighs that small ready-time advantage. The fusion credit helps `(p4, p5)` but does not make it competitive.

After applying the winner, the layout is `[q2, q1, q3, q4, q6, q5]`:

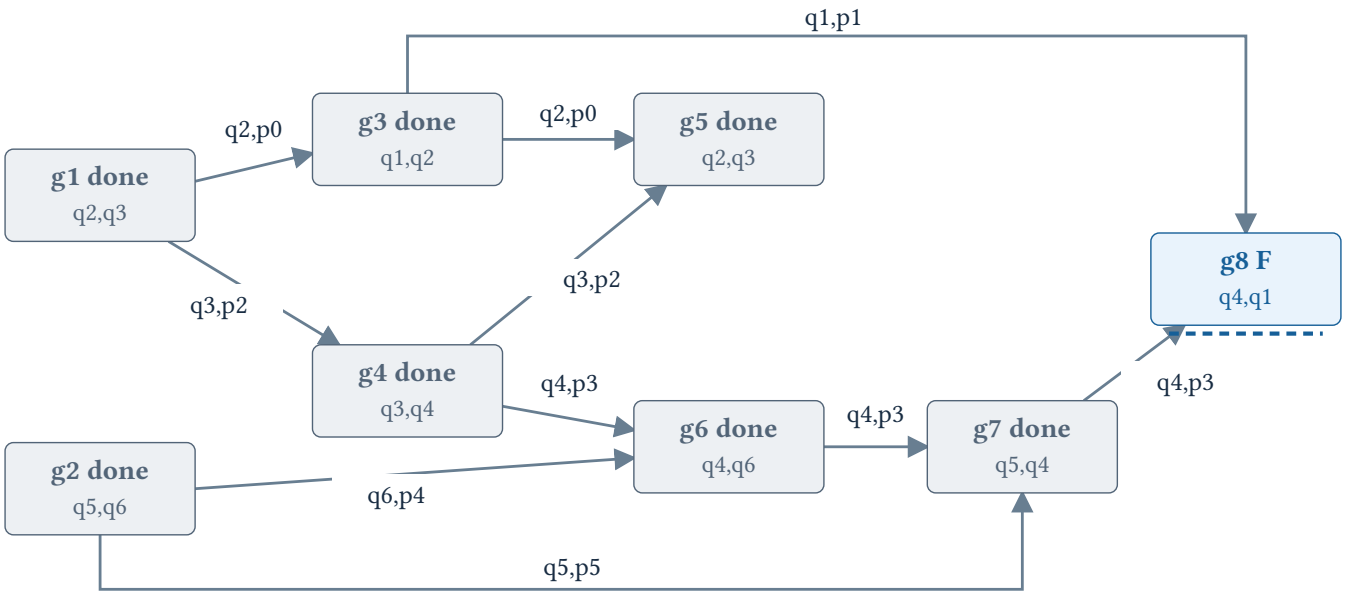
Decay variant / after decision 5

Decay chooses SWAP(p3,p4) at decision 5. SWAP(p1,p2) then reaches state S6.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	1	2	4	3
q2	1	0	2	3	5	4
q3	1	2	0	1	3	2
q4	2	3	1	0	2	1
q5	4	5	3	2	0	1
q6	3	4	2	1	1	0

Layers at this snapshot

F = {g8}

E = {}

Gate / physical operands / distance

g8: (p3,p1) d = 2

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 16: Decay variant after decision 5: F={g8}, E is empty, q4 has moved to p3 while q1 is still at p1, and their matrix distance is 2.

g8 is still at distance 2. Decision 6 then chooses `SWAP(p1, p2)` and emits g8 at distance 1. The route uses the same snapshots S0 through S4, then the snapshot above instead of S5, then S6. It has the same six SWAPs and final layout as section 7; the last two SWAPs occur in the opposite selection order. Those two SWAPs use disjoint wires, so changing their selection order alone does not demonstrate a depth reduction in this example.

8.7 Exactly when the memory is cleared

A **search** here is a run of chosen SWAPs trying to make at least one front gate executable. There can be several such searches in one routing trial. The normal decay bookkeeping is:

1. **Score and choose** using the current multipliers.
2. **Apply the chosen SWAP**, then advance the search-step counter.

3. If that counter reaches 5, **reset all multipliers to 1** and the counter to 0. Do not increment either endpoint on this step. Otherwise, add `0.001` to each of the chosen SWAP's endpoint multipliers.
4. If no front gate became adjacent, score the next set of candidates with these updated multipliers.
5. If a gate became adjacent, emit it and all other gates enabled by the emit sweep. Then **reset all multipliers to 1** before the next search. This reset does **not** reset the search-step counter.

Thus there are two separate pieces of state: the per-wire multipliers and a counter for the periodic reset. Emitting gates clears only the first. The counter counts heuristic-selected search SWAPs across the whole routing trial; it does not count emitted CX gates or the safety valve's forced SWAPs.

Here is the actual timeline for the decay route above. "All 1" means every wire's multiplier is 1; the counter column is its value **after** the event.

Event	Counter	Multipliers above 1 after the event	Why
Start	0	All 1	Initial state.
Decision 1: SWAP(p2, p3)	1	p2, p3: 1.001	Increment the chosen endpoints.
Emit g1 and g3	1	All 1	Emit sweep clears memory, not the counter.
Decision 2: SWAP(p3, p4)	2	p3, p4: 1.001	Increment.
Emit g2, g4, g6 and g5	2	All 1	Emit sweep reset.
Decision 3: SWAP(p3, p4)	3	p3, p4: 1.001	Increment.
Emit g7	3	All 1	Emit sweep reset.
Decision 4: SWAP(p0, p1)	4	p0, p1: 1.001	No gate emitted; this survives into decision 5.
Decision 5: SWAP(p3, p4)	0	All 1	Fifth search SWAP: periodic reset replaces increment.
Decision 6: SWAP(p1, p2)	1	p1, p2: 1.001	Increment.
Emit g8	1	All 1	Emit sweep reset; routing ends.

Notice the timing at decision 5: it is **scored with** p0 and p1 at 1.001, then the winner is applied, and only **then** does the fifth-step reset occur. Resetting before scoring would erase the very preference shown in section 8.5. Decision 6, by contrast, sees all multipliers at 1 even though decision 5 emitted no gate. Its two distance-improving candidates are separated by ready time: `(p1, p2)` has `T = 11`, while `(p2, p3)` has `T = 15` after the SWAP on `(p3, p4)`. This checkout therefore picks `(p1, p2)`.

8.8 How strong is the preference, and what does it guarantee?

With increment `0.001` and a reset every fifth search SWAP, an endpoint can accumulate at most four increments between periodic resets. Its multiplier is therefore at most `1.004`; gate emissions often reset it sooner. This configuration makes the reuse cost small: at most **0.4% of the positive quantity being multiplied**.

For example, a post-SWAP distance score of 2 on a reused wire becomes 2.002; it still beats a score of 3 on fresh wires. But decay is applied to **every** candidate, not only to exact ties. A fresh candidate scoring 2.001 would beat that 2.002 even though its distance score was slightly worse. Whether a distance advantage survives depends on the actual totals and multipliers; there is no general rule that distance always wins.

Decay encourages parallelism; it does not guarantee minimum depth, fewer SWAPs, or an optimal route. It remembers recent search steps, not elapsed hardware time. The name also does not mean the stored values gradually shrink: in this implementation they increase on use and jump back to 1 on reset. It has nothing to do with physical decoherence or qubit relaxation times.

To trace this in the code, start with `choose_best_swap` in `route.rs`: it adds the current total and multiplies by the maximum endpoint entry. Then read `swap_map_trial` in the same file for the increment, fifth-step reset, and post-emission reset. Section 13 combines this with all the other bookkeeping.

9. Ties, randomness, trials, and the safety valve

Ties

Scores within `1e-10` of the best are treated as tied. The router keeps all tied candidates and picks one with a seeded random generator. The same seed always gives the same route.

Multiple trials

`swap_map` runs the routing several times with different seeds (the `trials` argument; the default is Qiskit's default process count, roughly the number of physical CPU cores), possibly in parallel, and keeps the attempt with the **fewest inserted**

SWAPs. Only top-level SWAPs are counted; SWAPs inside control-flow bodies (section 19) are ignored. Among equal counts, the earliest trial wins. Depth is not measured when choosing:

Trial	SWAPs inserted	Depth (measured afterwards, not used)	Chosen?
0	5	12	No
1	4	15	**Yes**
2	4	11	No, ties trial 1 and comes later

So "best next SWAP" and "best complete route" use different criteria. The per-step score guides construction; the swap count picks the winner.

The safety valve

A lower score is a preference, not a guarantee that a gate becomes executable. After every SWAP the router checks adjacency directly. It can happen that many SWAPs in a row improve the score without ever making a front gate adjacent, for example by chasing lookahead or fusion gains around in circles.

To prevent that, each search has an **attempt limit** (the Python wrappers use `10 × number of qubits`). If more than that many SWAPs are chosen without emitting any gate, the router:

1. **Rolls back** every SWAP chosen since the last emitted gate, restoring the layout, the layer bookkeeping, and this checkout's history arrays. Already-emitted gates stay.
2. Picks the front gate whose qubits are **closest**.
3. Finds a **shortest path** between its two positions.
4. Inserts SWAPs along that path, moving both ends toward the middle, until the pair is adjacent, and emits the gate.

Example on a five-position line with the front gate CX(q1, q5):

```
Start:          [q1, q2, q3, q4, q5]     d(q1, q5) = 4
SWAP(p0, p1):   [q2, q1, q3, q4, q5]     d = 3
SWAP(p1, p2):   [q2, q3, q1, q4, q5]     d = 2
SWAP(p4, p3):   [q2, q3, q1, q5, q4]     d = 1  -> emit CX on (p2, p3)
```

The rollback is a compiler-side undo. Those SWAPs were never sent to hardware, so no inverse SWAPs appear in the output.

10. The whole loop, with every step named

The same algorithm as section 3, now with the names used in `route.rs`:

```
before routing (callers of swap_map):
    SabreDAG::from_dag: compress the circuit into routing nodes (section 18)
    RoutingTarget: compute the distance table from the coupling map

State::begin
    emit every initially executable gate; fill the front and lookahead layers
    advance_routed: ready times and fusion history for those initial gates

swap_map_trial:
    while the front layer is not empty:
        remember the output position                  (rollback point)
        until some front gate becomes adjacent:
            choose_best_swap                           (sections 6.1–6.3, 8, 11–13)
            apply it to the layout and layers          (section 6.4)
            advance_swap: ready times +3, clear block history (sections 11–12)
            update decay multipliers                   (section 8)
            if the attempt limit is exceeded:
                roll back; force_enable_closest_node   (section 9)
        update_route:
            record the chosen SWAPs and the gate(s) that became adjacent
            decrement successors; emit anything adjacent (section 6.5)
            rebuild the lookahead layer
            advance_routed: ready times and fusion history for emitted gates
        reset decay multipliers
    return the order of SWAPs and gates, plus the initial and final layouts
```

The router's long-lived state, for readers following the code:

Field of `State`	Remembers
`layout`	Who is where right now.
`layers`	The front and lookahead layers, as physical partner pairs.
`predecessors`	The readiness counters, one copy per layer.
`decay`	The per-position multipliers of section 8.
`ready`	Estimated time each position becomes free (section 12).
`last_block`	The most recent fusible two-qubit block on each position (section 11).
`ready_undo`	Enough history to roll back `ready` and `last_block`.
`swap_scores`, `best_swaps`	Scratch space for candidates and the tied best.
`rng`, `seed`	The seeded random generator.

Part III — What this checkout adds to the score

Classic SABRE scores a candidate SWAP by distances alone. Two things it cannot see:

- **A SWAP placed right after a CX on the same pair is cheaper than three CX.** A later optimization pass merges them, so the SWAP's real cost is lower.
- **A SWAP on busy wires has to wait**, while a SWAP on idle wires can run in parallel with other work. Distance says nothing about this.

This checkout adds a term for each. Both are small compared with a distance step, both are computed from the history of the current routing attempt, and both apply in every heuristic mode (`basic`, `lookahead`, `decay`). Part II's walkthrough already used them at decisions 3 to 6; this part shows the numbers.

11. Fusion credit: a SWAP right after a CX is cheaper

The idea in plain words

Every SWAP the router inserts is extra work for the device. Most devices have no native SWAP: it is built from **three CX gates**. Classic SABRE therefore treats every SWAP as costing the same, three CX gates' worth, no matter where it lands.

This checkout notices a discount. If the two wires a SWAP would use have **just** run a CX together, a later optimization pass can merge that CX and the SWAP into fewer gates. On that pair, the SWAP really costs less than three CX. The **fusion credit** is the router's estimate of that discount. It is subtracted from the candidate's score, so a SWAP that can merge with the gate placed just before it looks more attractive than an otherwise identical SWAP elsewhere.

The worked example this section follows

Everything below is shown on one concrete moment of the running route: **decision 3**, starting from snapshot S2 of section 7. At that moment:

- The last gate emitted was g6 = CX(q4, q6). q4 was at p3 and q6 at p4, so it ran as `CX(p3, p4)`, followed by the Z gate on q4.
- The front gate is g7 = CX(q5, q4), with q5 at p5 and q4 at p3, distance 2.
- The candidates are SWAP(p2, p3), SWAP(p3, p4) and SWAP(p4, p5).

Classic SABRE picks SWAP(p4, p5). This checkout picks SWAP(p3, p4), because that SWAP lands on the same two wires as g6 and can merge with it. Four figures walk through why:

1. Why a SWAP right after a CX is cheaper, drawn as circuits.
2. How each routing node gets a cost estimate before routing starts.
3. What each wire remembers as the route is built, event by event.
4. How the credit changes the scores at decision 3.

Why a SWAP right after a CX is cheaper

A SWAP on positions a and b is the same as three CX gates in a row:

```
SWAP(a, b) = CX(a, b); CX(b, a); CX(a, b)
```

A CX undoes itself: doing the same CX twice in a row is the same as doing nothing. So when g6 = `CX(p3, p4)` is followed by `SWAP(p3, p4)`, the CX already in the circuit and the first CX of the SWAP meet and cancel:

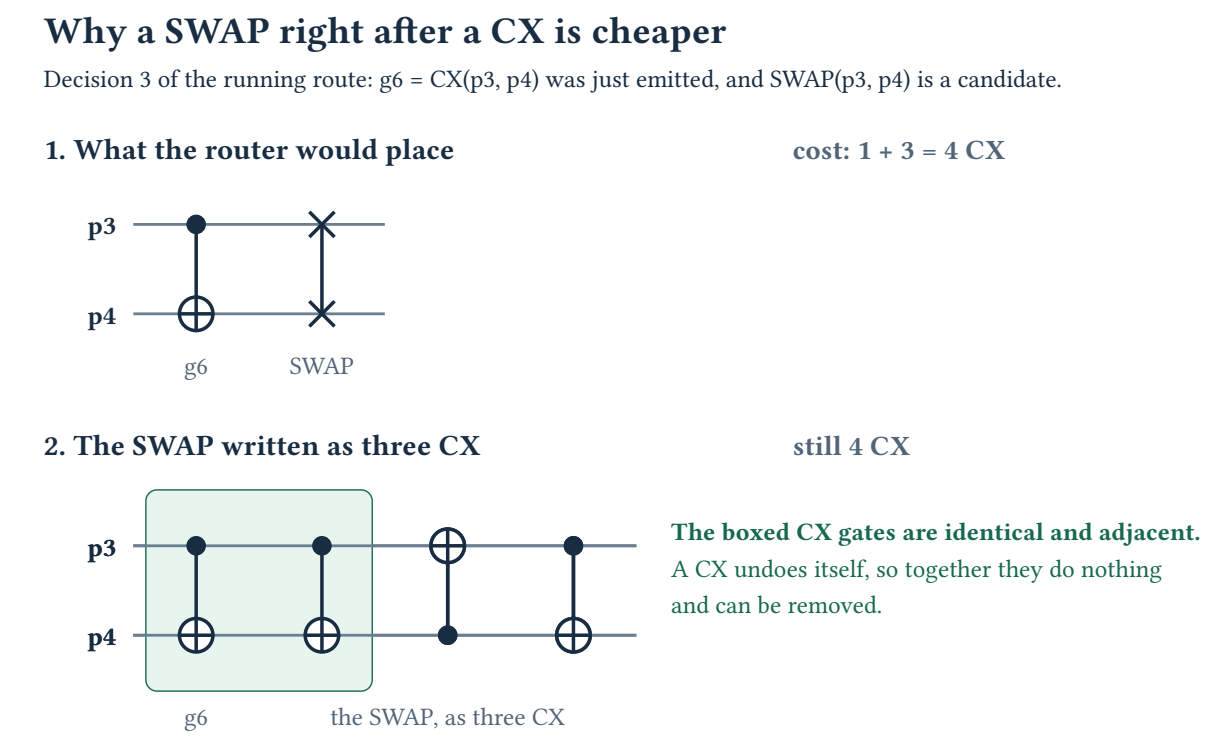


Figure 17: Why a SWAP right after a CX is cheaper. Panel 1: g6 = CX(p3, p4) followed by SWAP(p3, p4), 4 CX in total. Panel 2: the SWAP written as three CX; g6 and the SWAP's first CX are identical and adjacent, so they cancel. Panel 3: the two remaining CX, which have the same effect as g6 followed by the SWAP.

A CX followed by a SWAP therefore needs only **2 CX gates**, not 4. Throughout this section "cost" counts two-qubit gates only, in CX units. One-qubit gates such as H, Z or Rz are cheap and are not counted. (The real route has a Z gate on p3 between g6 and the SWAP. It is a one-qubit gate on the control wire, and it slides past the CX without changing anything, so the count above is unchanged.)

It still works with one-qubit gates in the way. Read each sequence from left to right. The vertical bar separates the gates already in the circuit from the SWAP being appended; it is only a separator in this explanation, not a circuit barrier. Example 1: two CX gates, then a SWAP.

```
Already in the circuit          | Appended SWAP
CX(b,a); H(a); CX(a,b)          | CX(a,b); CX(b,a); CX(a,b)
```

Counted separately this is `2 + 3 = 5` CX gates. But the last `CX(a,b)` on the left and the first `CX(a,b)` on the right are identical and adjacent, so they cancel:

```
CX(b,a); H(a); CX(b,a); CX(a,b)                3 CX + 1 H
```

This shorter sequence does exactly the same thing to the two qubits as the original gates followed by the SWAP. The SWAP's effect is now inside those three CX gates; there is no separate SWAP left to run.

Example 2: three CX gates, then a SWAP.

```
Already in the circuit                | Appended SWAP
CX(a,b); H(a); CX(b,a); CX(a,b)       | CX(a,b); CX(b,a); CX(a,b)
```

Counted separately, `3 + 3 = 6`. Cancel the two `CX(a,b)` gates at the separator, and then the two `CX(b,a)` gates that become adjacent:

```
CX(a,b); H(a); CX(b,a); CX(b,a); CX(a,b)       4 CX + 1 H
CX(a,b); H(a); CX(a,b)                         2 CX + 1 H
```

Here the merged result has **fewer** CX gates than the original run alone.

Any run of gates on one pair needs at most three CX. The cancellations above are cases you can see by eye. There is also a general rule that does not depend on spotting cancellations: *a run of gates that all act on the same two qubits, however long, can always be rewritten as at most three CX gates plus one-qubit gates.* The reason, in one sentence, is that such a run does one fixed thing to those two qubits, and there is a standard recipe that builds any such action from three CX gates and a few one-qubit rotations. The proof is in Vidal and Dawson, Theorem 1; the mathematics is outside the scope of this guide. Qiskit's optimization passes apply this rewriting to two-qubit blocks after routing.

The rule has one condition: **everything in the run must stay on the same two qubits.** A gate that touches a third qubit, such as `CX(b, d)`, ends the run. So do measurements, resets, barriers, and other non-gate instructions.

Two consequences for the router:

- Appending a SWAP to a run that already holds one CX gives 2 CX, as the cancellation shows.
- Appending a SWAP to a run that already holds two or three CX gives at most 3 CX, by the general rule. A run that is already at three CX absorbs the SWAP for free.

So the router never needs to count a run beyond three CX. That is why every estimate below is **capped at 3**.

What the router remembers

The router does not inspect gate sequences while it scores candidates; that would be far too slow. Instead it keeps a small amount of bookkeeping, in two steps: one before routing starts, one as gates are emitted.

Step 1, before routing: each routing node gets a block estimate. When the circuit is compressed for routing (section 18), consecutive operations that need the same two program qubits to be adjacent are grouped into one **routing node**. Each node carries a numeric field, **`block`**: a rough estimate of how many CX gates the node adds to a run on its pair.

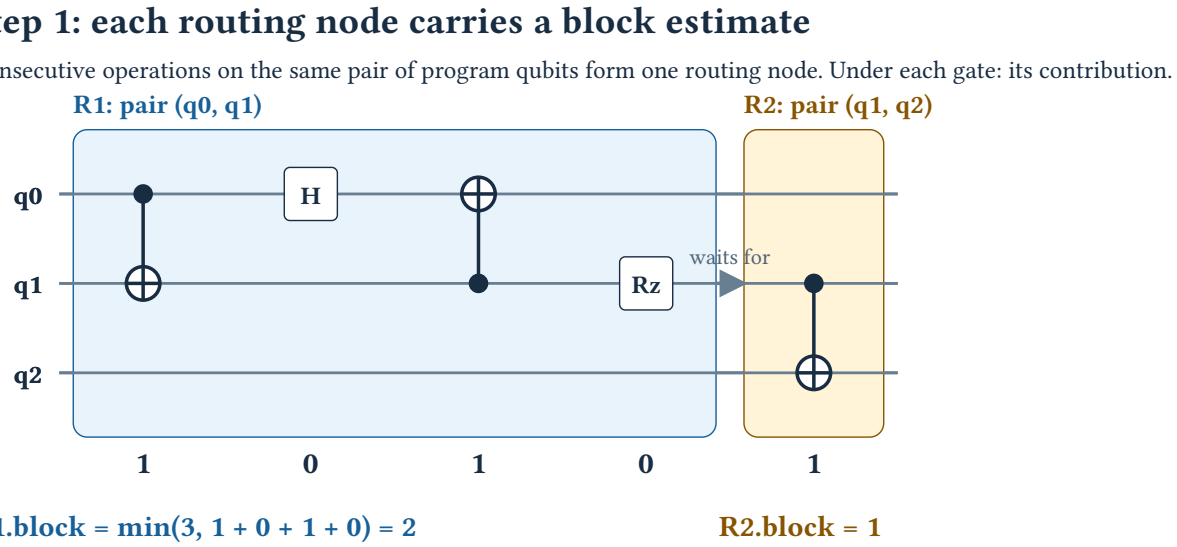


Figure 18: Step 1: each routing node carries a block estimate. A circuit on q0, q1, q2 with CX(q0, q1), H on q0, CX(q1, q0) and Rz on q1 grouped into node R1 for the pair (q0, q1), followed by CX(q1, q2) as node R2. The contributions 1, 0, 1, 0 give R1.block = 2; R2.block = 1.

In the figure, once q0 and q1 sit on adjacent positions, all four operations of R1 can run without another routing decision, so they form one node. The next operation, `CX(q1, q2)`, needs a different pair and becomes node R2, which must wait for

R1. Grouping does not turn the operations into one hardware gate: when R1 is emitted, its four operations are emitted in order.

The estimate adds up a fixed contribution per operation and caps the total at 3:

Operation in the node	**Contribution to `node.block`**
CX, CY, CZ, ECR	1
Any other two-qubit gate	3 (worst case)
One-qubit gate folded into the node	0
Measurement, reset, barrier, delay, control flow, and other non-gate instructions	none: the node's `block` becomes `None`, meaning "this node ends any run"

The cap means a node with four CX, or ten, stores 3 just like a node with three. The original gates are kept; the cap only limits the estimate. The router does not look for cancellations here, so Example 2's eventual two-CX result is not predicted. The estimate is deliberately coarse.

In the running example every two-qubit node is a single CX, sometimes with one-qubit gates folded in (g4 carries two H gates, g6 carries a Z), so every node has `block = 1`.

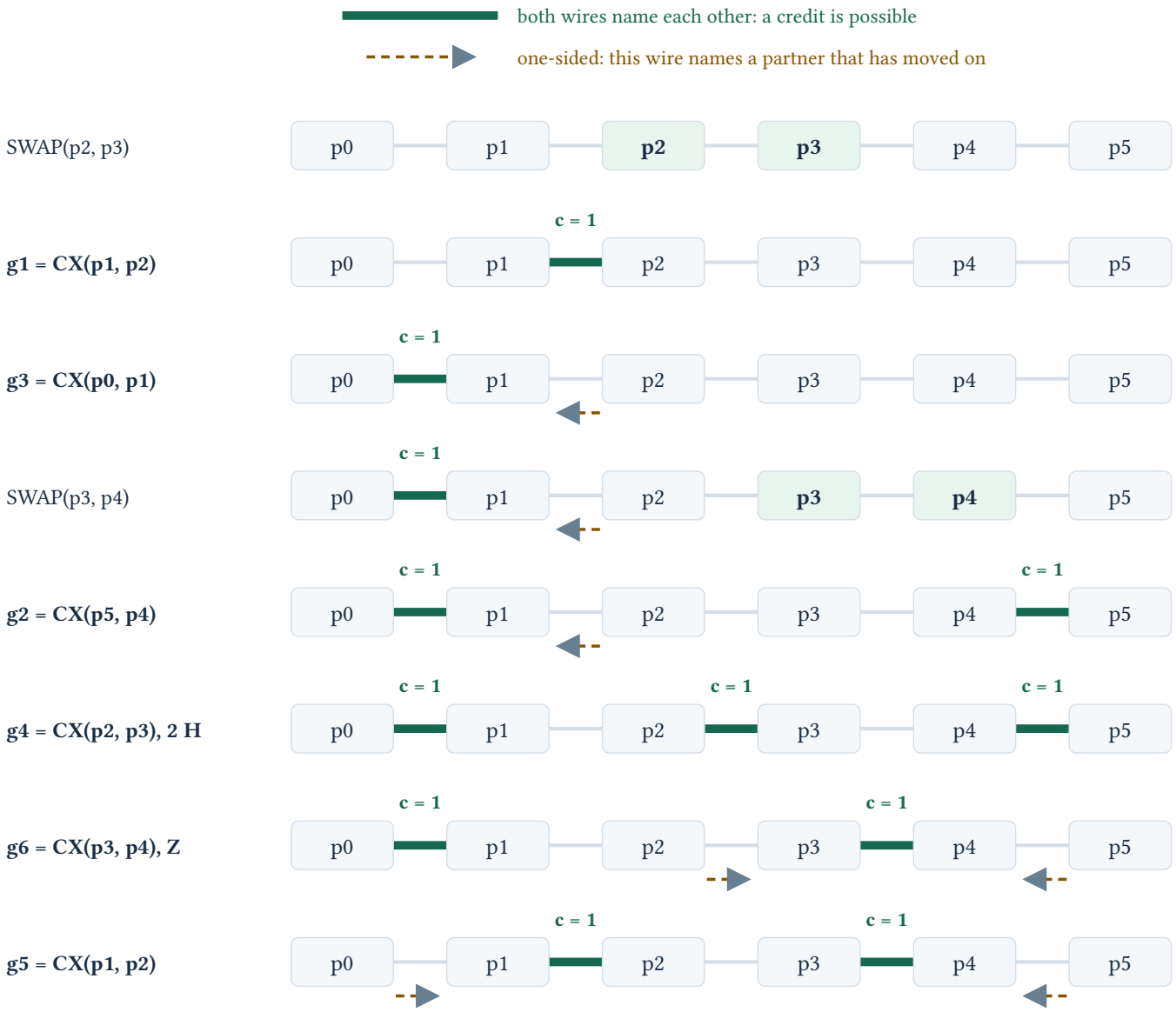
Step 2, during routing: each physical position remembers its latest partner. For every physical position p the router keeps

```
last_block[p] = (partner, c)
```

which reads: "the last two-qubit work on wire p was with wire `partner`, and that run is worth about `c` CX gates (at most 3)". Here `c` is the running total for the current run on that pair. The next figure replays the running route from decision 1 up to decision 3 and shows these records after every event. A solid green link means the two wires name each other. A dashed arrow means a wire still names a partner that has since moved on to someone else.

Step 2: what each wire remembers, event by event

The running route from decision 1 up to decision 3. After each event: which wire names which, and the run cost c.



A SWAP clears both of its wires. A gate on (a, b) makes a and b name each other; any older record on a or b is overwritten.

Figure 19: Step 2: what each wire remembers, event by event. Eight strips, one per event from SWAP(p2, p3) through g5 = CX(p1, p2), each showing the six wires p0 to p5 with green links where both wires name each other and dashed arrows for one-sided records. After the last event, p1 and p2 name each other and p3 and p4 name each other, while p0 still names p1 and p5 still names p4.

Read the strips from top to bottom:

- **SWAP(p2, p3).** The first decision. Nothing has been emitted yet, so there is nothing to clear.
- **g1 = CX(p1, p2).** The first gate on the pair (p1, p2): p1 and p2 now name each other with `c = 1`.
- **g3 = CX(p0, p1).** p0 and p1 name each other. p1′s old record, which named p2, is overwritten. p2 still names p1, but p1 no longer names p2: the dashed arrow. A SWAP on (p1, p2) could no longer merge with g1, because g1 is no longer the latest thing on p1.
- **SWAP(p3, p4).** Decision 2. Both wires are cleared; they were empty anyway.
- **g2, g4, g6, g5.** Each gate makes its two wires name each other. Watch g6 = CX(p3, p4): p3 used to name p2 and p4 used to name p5, so this is a fresh run with `c = 1`, and p2 and p5 are left with one-sided records.

The rules the figure follows:

Event	Effect on `last_block`
A two-qubit node is emitted on (a, b)	If a and b already name each other, the run continues: `c = min(3, old c + node.block)`. Otherwise a fresh run starts with `c = node.block`. Both a and b now name each other. If the node's `block` is `None`, both records are cleared instead.
A SWAP is chosen on (a, b)	Both records are cleared. The credit for that run has been used; the next gate on the pair starts a new run.
Any other node runs on p (measurement, reset, barrier, delay, control flow, an operation on three or more qubits)	p's record is cleared (final routing only; see section 17).

The "run continues" case does not occur in the running example, because no two consecutive nodes land on the same pair of wires. It would look like this: if R1 from the Step 1 figure were emitted on (p2, p3) with no earlier run there, the record would be `c = 2`; if another node with `block = 1` on the same pair followed, `c` would become `min(3, 2 + 1) = 3`.

Two quantities are easy to mix up, so here they are side by side:

Quantity	What it describes	When it is computed
`node.block`	One routing node's contribution, from its original operations.	When the circuit is compressed, before routing.
`c`	The accumulated estimate for the current run of emitted gates on one physical pair.	During routing, as nodes are emitted.

The handshake. A candidate `SWAP(a, b)` qualifies for a credit only when `last_block[a]` names b **and** `last_block[b]` names a: a solid green link in the figure, never a dashed arrow. Both must agree because a wire can move on, as the g3 strip shows. The one-sided record correctly earns nothing.

Gates that are emitted at the very start, before any SWAP, follow the same rules: the routine that applies them (`advance_routed`) runs for the initially executable gates too, and likewise for the ready times of section 12. Section 16 relies on this.

The credit, in numbers

When a candidate passes the handshake with run cost `c`, the router compares two estimates:

```
without fusion:  c + 3          (the run, then a full three-CX SWAP)
with fusion:     min(3, c + 1)  (the run and the SWAP merged)
saving         = (c + 3) − min(3, c + 1)
credit         = −saving / 3
```

The merged estimate `min(3, c + 1)` is just the two consequences from earlier written as one formula: `c + 1` gives 2 when `c = 1` (the cancellation), and the cap at 3 covers `c = 2` and `c = 3` (the general rule).

The division by 3 converts gates into the score's units. The distance terms of sections 6 and 8 count SWAPs, and one SWAP is three CX, so a saving of two CX is two thirds of a SWAP.

Run cost c	Separate cost	Merged estimate	Saving	Credit
no qualifying run	—	—	0	0
1 (one CX)	4	2	2	−2/3
2	5	3	2	−2/3
3	6	3	3	−1

A credit of −2/3 is worth two thirds of a distance step. That is deliberately large: it is the estimated saving in real gates, not a tie-breaker.

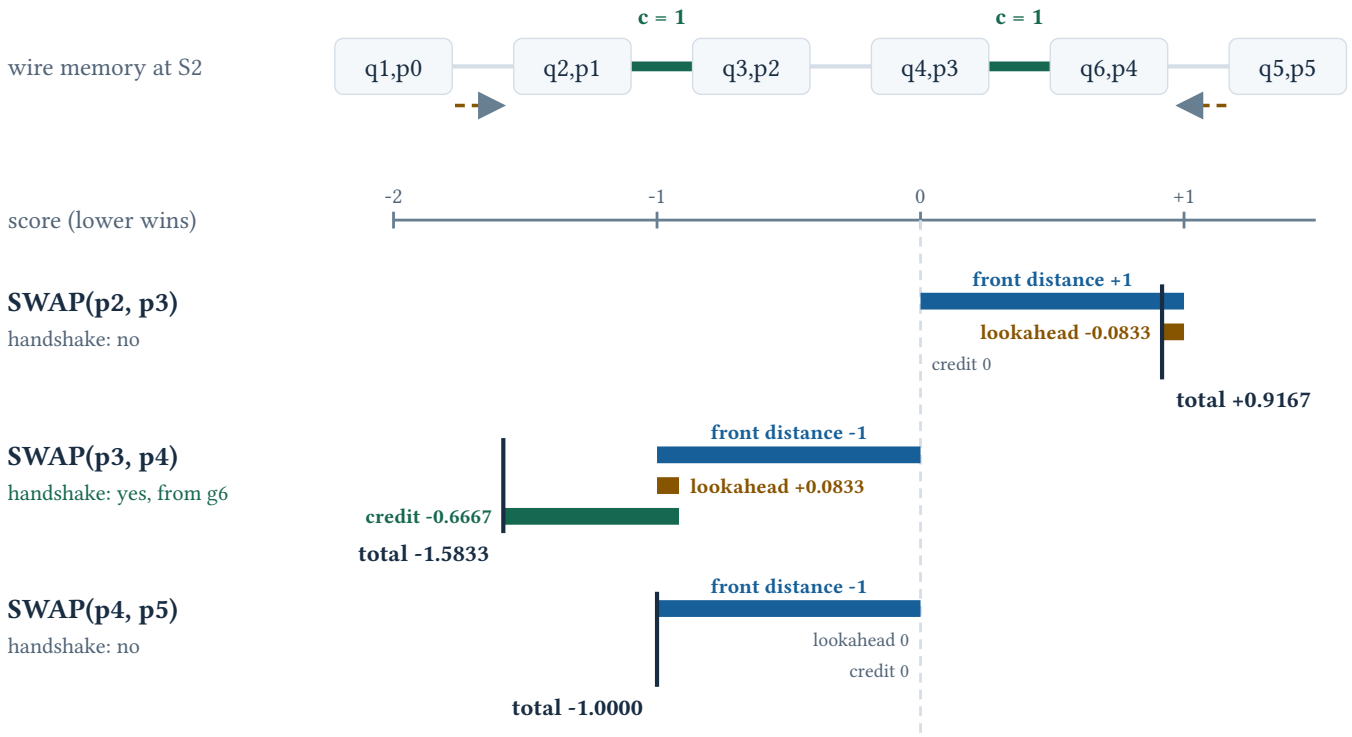
This is a routing estimate, not a promise about the final gate count. The router still emits a complete three-CX SWAP. Whether it is merged later is up to Qiskit's optimization passes; the router only predicts the saving and uses it to choose.

Decision 3, scored

Now put the pieces together. The last strip of the history figure is the wire memory at S2: p1 and p2 name each other, p3 and p4 name each other, and p0 and p5 hold one-sided records. Of the three candidates, only SWAP(p3, p4) passes the handshake, thanks to g6. Its run cost is `c = 1`, so its credit is −2/3.

Decision 3: the credit overturns the classic ranking

Top: wire memory at S2 (the last strip of the previous figure). Bottom: each candidate's score, one term at a time.



Without the credit SWAP(p4, p5) wins by 1/12. The credit of -2/3 is eight times larger and moves SWAP(p3, p4) to the front.
A credit never beats a full distance step: a SWAP that pushes a front pair apart still scores +1/3 at best.

Figure 20: Decision 3: the credit overturns the classic ranking. Top: the wire memory at S2. Bottom: for each candidate, the front-distance change, the lookahead term and the credit laid end to end on a number line from −2 to +1. SWAP(p2, p3) totals +0.9167, SWAP(p3, p4) totals −1.5833 and wins, SWAP(p4, p5) totals −1.

The same numbers as a table:

Candidate	ΔD_F	w × ΔD_E	Handshake?	Credit	Sum
SWAP(p2, p3)	+1	−0.0833	No: p2 names p1, p3 names p4	0	+0.9167
SWAP(p3, p4)	−1	+0.0833	Yes, from g6	−2/3	**−1.5833**
SWAP(p4, p5)	−1	0	No: p4 names p3	0	−1.0000

Read the figure row by row. SWAP(p2, p3) moves q4 away from q5, so its front-distance term is +1 and it is out of the running. SWAP(p3, p4) and SWAP(p4, p5) both bring g7′s qubits together, so both start at −1. Classic SABRE then separates them only by the lookahead: SWAP(p3, p4) moves q4 one step further from q1, which g8 will need, and pays one twelfth for it. On distances alone SWAP(p4, p5) wins by that twelfth. The credit of −2/3 is eight times larger and pulls SWAP(p3, p4) well in front.

This is not a tie-break. The credit can beat the lookahead term, whose weight is at most `w` per gate. It cannot beat a full distance step in the wrong direction: a SWAP that moves a front pair *apart* (ΔD_F = +1) still scores +1/3 with the credit.

Decisions 1 and 2 had no credit: no two-qubit gate had yet been emitted on any candidate pair, as the first and fourth strips of the history figure show.

Decisions 4 to 6 show the limit. After SWAP(p3, p4) is chosen, p3 and p4 are cleared. g7 = CX on (p5, p4) then makes p4 and p5 name each other, so SWAP(p4, p5) carries a credit of −2/3 at decisions 4, 5 and 6. It never wins, because it moves q4 away from q1: its score is +1 − 2/3 = +1/3 each time, against −1 for the candidates that close the distance.

What the estimate does not check

The bookkeeping is coarse on purpose, so that it costs almost nothing per candidate. It does not verify that the later optimization pass will really merge the run and the SWAP: gate parameters, opaque instructions, and the optimization

level all matter, so a credited saving can fail to materialize. That affects route quality only, never correctness. The router always emits a complete SWAP, and whether it gets merged later is up to other passes.

12. Ready time: prefer wires that are free sooner

The idea

Think of each physical position as having a clock: "the two-qubit work already assigned to this wire ends at time T". A gate on two wires can start only when both are free. If two candidate SWAPs are otherwise equal, the one whose wires free up earlier can start earlier and is less likely to lengthen the circuit.

The bookkeeping

`ready[p]` starts at 0 for every position and is updated like this:

```
When a SWAP is chosen on (a, b):          end = max(ready[a], ready[b]) + 3
                                          ready[a] = ready[b] = end

When a two-qubit node is emitted on (a, b): end = max(ready[a], ready[b]) + 1
                                          ready[a] = ready[b] = end
```

A SWAP counts as 3 layers (three native gates); a routed two-qubit node counts as 1, even if it contains several gates. Ready times belong to **positions**, not to program qubits, so they are not exchanged by a SWAP.

Through Part II's route:

Event	p0	p1	p2	p3	p4	p5
Start	0	0	0	0	0	0
SWAP(p2, p3)	0	0	3	3	0	0
g1 on (p1, p2)	0	4	4	3	0	0
g3 on (p0, p1)	5	5	4	3	0	0
SWAP(p3, p4)	5	5	4	6	6	0
g2 on (p5, p4)	5	5	4	6	7	7
g4 on (p2, p3)	5	5	7	7	7	7
g6 on (p3, p4)	5	5	7	8	8	7
g5 on (p1, p2)	5	8	8	8	8	7
SWAP(p3, p4)	5	8	8	11	11	7
g7 on (p5, p4)	5	8	8	11	12	12
SWAP(p0, p1)	11	11	8	11	12	12
SWAP(p1, p2)	11	14	14	11	12	12
SWAP(p3, p4)	11	14	14	15	15	12
g8 on (p3, p2)	11	14	16	16	15	12

Read the g2 and g4 rows: g2 on (p5, p4) starts at `max(0, 6) = 6`, right after the SWAP on p4, and ends at 7; g4 on (p2, p3) also starts at `max(4, 6) = 6` and ends at 7. The estimate lets them overlap, because they use different wires. The one-qubit gates folded into g4 and g6 add nothing: a routed node counts as one layer whatever it contains.

The penalty

Each candidate SWAP on (a, b) pays

```
ready penalty = 0.000022 × max(ready[a], ready[b])
```

The coefficient is tiny, so this term only matters when everything else is nearly equal, but it is added to every candidate, not applied only after exact ties. A large enough difference in ready time can outweigh a small lookahead difference.

Decision 3 in full (lookahead mode, so no decay):

Candidate	ΔD_F	w × ΔD_E	Credit	Ready of endpoints	Penalty	Total
SWAP(p2, p3)	+1	−0.0833	0	8, 8	0.000176	+0.916843
SWAP(p3, p4)	−1	+0.0833	−2/3	8, 8	0.000176	**−1.583157**
SWAP(p4, p5)	−1	0	0	8, 7	0.000176	−0.999824

Here the credit already decided; the ready penalties are equal and change nothing.

Decisions 4 to 6: the ready term decides by itself

After g7, only g8 = CX(q4, q1) is left, at distance 4, and the lookahead layer is empty. At each of the next three decisions two candidates close the distance by one step and nothing but the ready term separates them. The ready times are those of the table above.

Decision 4 (`ready = [5, 8, 8, 11, 12, 12]`):

Candidate	ΔD_F	Credit	Ready of endpoints	Penalty	Total
SWAP(p0, p1)	−1	0	5, 8	0.000176	**−0.999824**
SWAP(p3, p4)	−1	0	11, 12	0.000264	−0.999736
SWAP(p4, p5)	+1	−2/3	12, 12	0.000264	+0.333597

Decision 5 (`ready = [11, 11, 8, 11, 12, 12]`):

Candidate	ΔD_F	Credit	Ready of endpoints	Penalty	Total
SWAP(p1, p2)	−1	0	11, 8	0.000242	**−0.999758**
SWAP(p3, p4)	−1	0	11, 12	0.000264	−0.999736
SWAP(p0, p1)	+1	0	11, 11	0.000242	+1.000242
SWAP(p4, p5)	+1	−2/3	12, 12	0.000264	+0.333597

Decision 6 (`ready = [11, 14, 14, 11, 12, 12]`):

Candidate	ΔD_F	Credit	Ready of endpoints	Penalty	Total
SWAP(p3, p4)	−1	0	11, 12	0.000264	**−0.999736**
SWAP(p2, p3)	−1	0	14, 11	0.000308	−0.999692
SWAP(p1, p2)	+1	0	14, 14	0.000308	+1.000308
SWAP(p4, p5)	+1	−2/3	12, 12	0.000264	+0.333597

The margins are a few hundred-thousandths, but they are systematic: the candidate whose wires have been idle longest wins. Classic SABRE would flip three coins here. The result is that q1 walks two steps along the wires p0, p1, p2, which have been free since g3 and g5, while the busy wires p3 to p5 are still finishing g7, and only the last step uses p3 and p4. Running this in Qiskit (section 21), every seed gives the same output circuit.

What the number means

`ready[p]` is an estimated layer index counted from the start of the circuit. It is not seconds, not an error rate, and not a measured idle time. It is a cheap proxy that steers the router toward parallel-friendly choices.

13. The complete score, in the order the code computes it

Symbol	Meaning
`ΔD_F`	Change in the summed front-layer distances (section 6.2).
`ΔD_E`	Change in the summed lookahead-layer distances (section 6.3).
`b`	Basic weight: 1, or `1 / (number of front gates)` in `basic` mode.
`w`	Lookahead weight: `0.5 / (number of physical qubits)`.
`f`	Fusion credit: 0, −2/3, or −1 (section 11).
`H_now`	Current weighted distance total, `b × D_F + w × D_E` (section 8).
`m`	Larger decay multiplier of the two endpoints (section 8).
`T`	Larger ready time of the two endpoints (section 12).

`choose_best_swap` computes, for every candidate:

```
x = b × (f + ΔD_F)              fusion credit and front distance, scaled together
x = x + w × ΔD_E                lookahead
if decay is enabled:
    x = (H_now + x) × m         make it absolute, then penalize reused wires
score = x + 0.000022 × T        ready-time penalty, added last and never multiplied
```

The candidate with the smallest score wins; ties within `1e-10` are broken at random.

Two details that are easy to miss: the basic weight scales the fusion credit as well as the distance change, and the ready penalty is added *after* decay, so the decay multiplier never touches it.

The bookkeeping, in event order

To route a new circuit by hand with this checkout's exact choices, update the state in this order:

1. **Choose** the candidate with the smallest score (formula above).
2. **Apply the SWAP** on (a, b): swap the occupants of a and b in the layout.
3. **Ready times:** `ready[a] = ready[b] = max(ready[a], ready[b]) + 3`.
4. **Fusion history:** clear `last_block[a]` and `last_block[b]`.
5. **Decay** (decay mode only): add 0.001 to a's and b's multipliers, or reset all multipliers if this is the fifth search SWAP.
6. **Check adjacency** of the front gates on a and b. If none is adjacent, go back to step 1 (the safety valve of section 9 bounds this).
7. **Emit** each newly adjacent gate on its positions (x, y), then any successor that becomes ready and is adjacent, and so on. For each emitted two-qubit node: `ready[x] = ready[y] = max(ready[x], ready[y]) + 1`, and update `last_block[x]`, `last_block[y]` by the rules of section 11. Blocked successors join the front layer.
8. **Rebuild the lookahead layer** from the new front; reset all decay multipliers to 1; go back to step 1 if the front is not empty.

The three named modes of the Python wrappers:

Mode	`b`	Lookahead	Decay	Fusion and ready terms
`basic`	`1 / number of front gates`	none	none	included
`lookahead`	1	one layer, `w = 0.5 / qubits`	none	included
`decay`	1	same as lookahead	+0.001 per use; every fifth search SWAP resets all to 1	included

`SabreSwap` defaults to `basic`; `SabreLayout`'s integrated path uses the `decay` configuration. In `basic` mode the two-gate improvement of −2 from section 6.2 would be reported as −1, because it is averaged over the two front gates.

14. What these estimates do not know

Collected here so that the main text can stay simple:

- A compressed routing node (section 18) adds **1** to ready time even if it holds several two-qubit gates, while its fusion estimate may be 3. The two fields are different approximations.
- Barriers and measurements clear fusion history but do **not** synchronize ready times. After a global barrier, `[10, 10, 0, 0]` stays as it is, even though later work on p2 and p3 really has to wait.
- A chosen SWAP adds 3 to ready time even when the fusion credit predicted it would merge into a cheaper block.
- Nested control-flow blocks are routed with their own fresh state; the outer ready times do not include their duration.
- Complete routing attempts are still compared by **top-level SWAP count** (section 9), not by native gate count or depth. The new terms change how each attempt is built, not how the winner is picked.
- Both terms are history, so the safety valve (section 9) rolls them back together with the layout when it undoes a failed search.

Part IV — Choosing the initial layout

15. The idea: route it backward

Part II started with everyone in a fixed arrangement and paid two SWAPs before the first two gates could run, then three more at the end to bring q4 and q1 together. If q6 had started next to q5, and q1 nearer to q4, several of those SWAPs would have been unnecessary. `SabreLayout` searches for a starting arrangement that suits the *whole* circuit, not just its first gates, and it does so by reusing the router.

The trick has two observations:

1. **The end of a forward route is a good arrangement for the end of the circuit.** Routing moves qubits toward the pairs that interact; by the last gate, the layout has adapted to the circuit's late interactions.
2. **Routing the dependencies in reverse carries that arrangement back to the start.** Reverse every "must come before" arrow, so the last gates come first, and route again starting from the forward route's final layout. The final layout of *that* route is adapted to the circuit's first gates, while still remembering what the end needed.

```
starting guess L0
    ── route forward ──▶ final layout L1      (good for the end of the circuit)
    ── route backward ─▶ final layout L2      (good for the start, informed by the end)

use L2 as the initial layout, or as the next starting guess
```

"Backward" means the **dependency graph is reversed**: g5 and g8 are visited first and g1 and g2 last. The compiler is not running the circuit in reverse time, inverting gates, or undoing measurements. Only the pattern of *which pairs interact in which order* matters for choosing a layout, and that pattern can be walked in either direction.

16. Worked example: six SWAPs become four

`SabreLayout` runs every pass with the decay configuration (section 13), so the numbers below include the decay multiplier; everything else is the scoring of Parts II and III. Each pass starts with fresh ready times, fusion history and decay multipliers; only the layout carries over.

Forward pass. Route the circuit from `[q1, q2, q6, q3, q4, q5]`. This is Part II's route again with one change: decay reverses the order of the last two SWAPs (section 8 works out that decision). The pass ends at `[q2, q3, q1, q4, q6, q5]`. Keep only that final layout.

Backward pass: states B0 to B5

Reverse every dependency, so the obligations now run g5 and g8 first, then g3 and g7, and so on back to g1 and g2. Start from `[q2, q3, q1, q4, q6, q5]`:

Start, B0. The layout carries over, but the completed-gate set starts empty for this new pass. g5 on (p0,p1) and g8 on (p3,p2) are adjacent and can be visited immediately. This makes g3 and g7 ready, both at distance 2. `F = {g3,g7}`, `E = {g6}`; g4 still waits for g6.

The graph uses the **same node positions as the forward walkthrough, with every arrow reversed**. Grey now means visited in this backward pass. These visits explore a layout; they do not append gates to the final output circuit. The distance matrix uses the same fixed logical-qubit order as before.

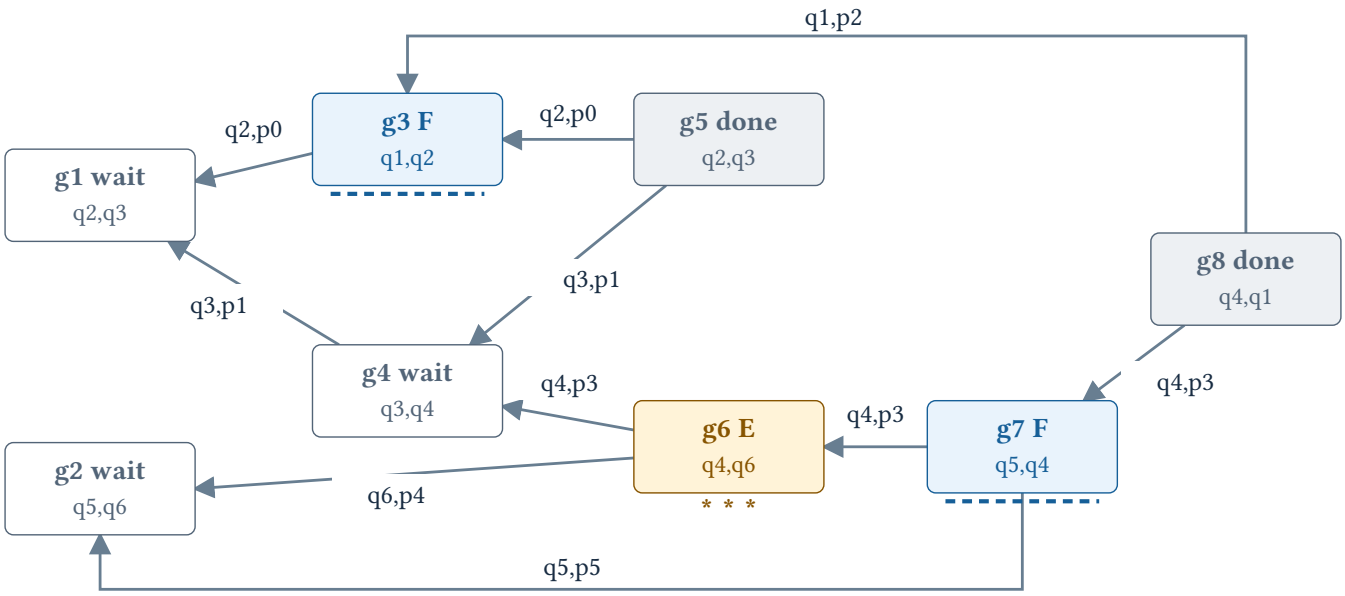
Backward state B0

Before B1: visit g5 and g8. Arrows now point toward earlier circuit gates.

Physical line / current occupants



Dependency graph / reversed arrows



--- F: front *** E: lookahead done: already visited wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	2	1	1	3	2
q2	2	0	1	3	5	4
q3	1	1	0	2	4	3
q4	1	3	2	0	2	1
q5	3	5	4	2	0	1
q6	2	4	3	1	1	0

Layers at this snapshot

F = {g3, g7}

E = {g6}

Gate / physical operands / distance

g3: (p2,p0) d = 2

g7: (p5,p3) d = 2

g6: (p3,p4) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 21: Backward state B0, before B1: g5 and g8 have been visited; reversed dependencies give F={g3,g7} and E={g6}.

Decision B1. SWAP(p0,p1) brings q2 next to q1 and carries a fusion credit from g5 on (p0,p1). Visit g3 on (p2,p1). g1 still waits for g4.

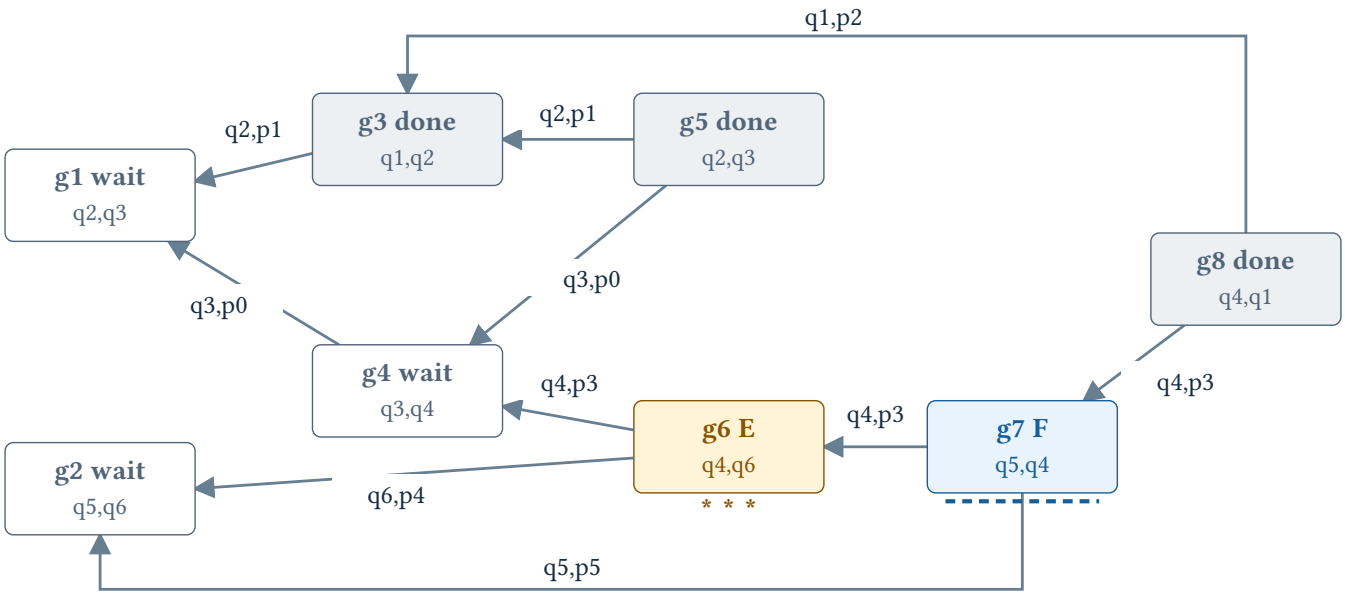
Backward state B1

After B1: visit g3. g1 still waits for g4. Next: B2.

Physical line / current occupants



Dependency graph / reversed arrows



--- F: front *** E: lookahead done: already visited wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	2	1	3	2
q2	1	0	1	2	4	3
q3	2	1	0	3	5	4
q4	1	2	3	0	2	1
q5	3	4	5	2	0	1
q6	2	3	4	1	1	0

Layers at this snapshot

F = {g7}

E = {g6}

Gate / physical operands / distance

g7: (p5,p3) d = 2

g6: (p3,p4) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 22: Backward state B1, before B2: q2 is now at p1, g3 is done, F={g7}, and E={g6}.

Decision B2. SWAP(p3,p4) makes g7 adjacent. Lookahead prefers it to SWAP(p4,p5), which would move q6 away from q4. Visit g7 on (p5,p4), then g6 on (p4,p3). g4 and g2 become the front, at distances 4 and 2.

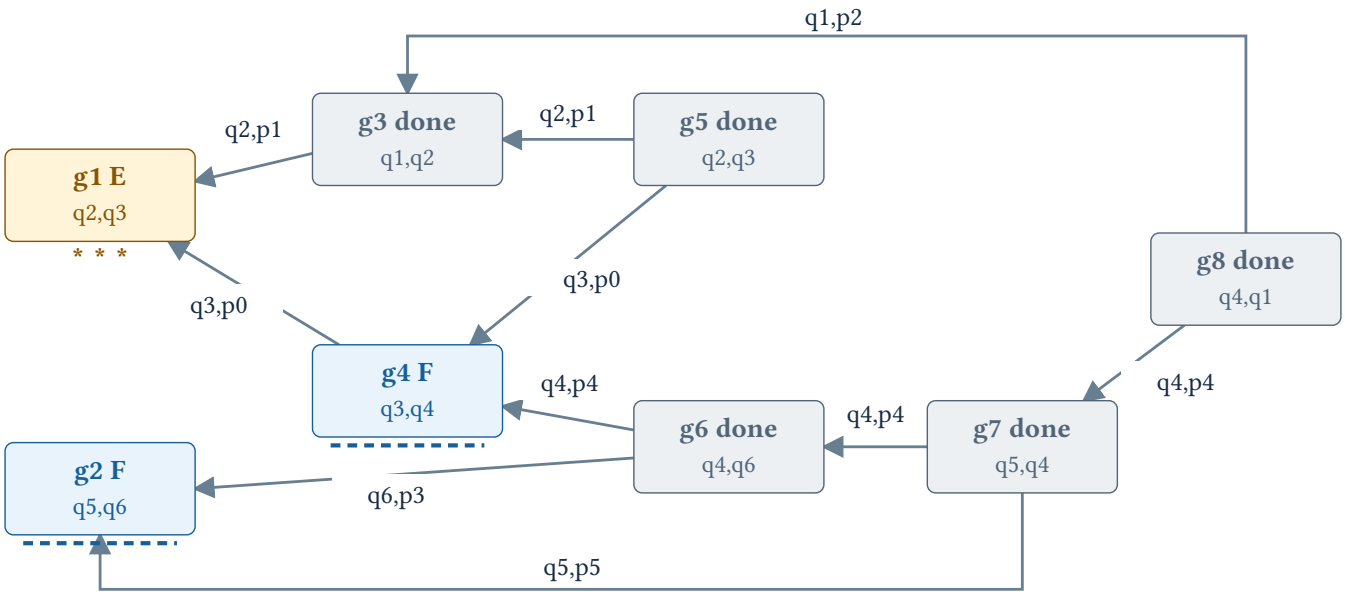
Backward state B2

After B2: visit g7, then g6. Next: B3.

Physical line / current occupants



Dependency graph / reversed arrows



--- F: front *** E: lookahead done: already visited wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	2	2	3	1
q2	1	0	1	3	4	2
q3	2	1	0	4	5	3
q4	2	3	4	0	1	1
q5	3	4	5	1	0	2
q6	1	2	3	1	2	0

Layers at this snapshot

F = {g2, g4}

E = {g1}

Gate / physical operands / distance

g2: (p5,p3) d = 2

g4: (p0,p4) d = 4

g1: (p1,p0) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 23: Backward state B2, before B3: g7 and g6 are done; F={g2,g4}, E={g1}. The matrix shows d(q3,q4)=4 and d(q5,q6)=2.

Decision B3. SWAP(p3,p4) has `ΔD_F = -2`, helping g4 and g2 at once, plus a fusion credit from g6. Visit g2 on (p5,p4). g4 remains at distance 3.

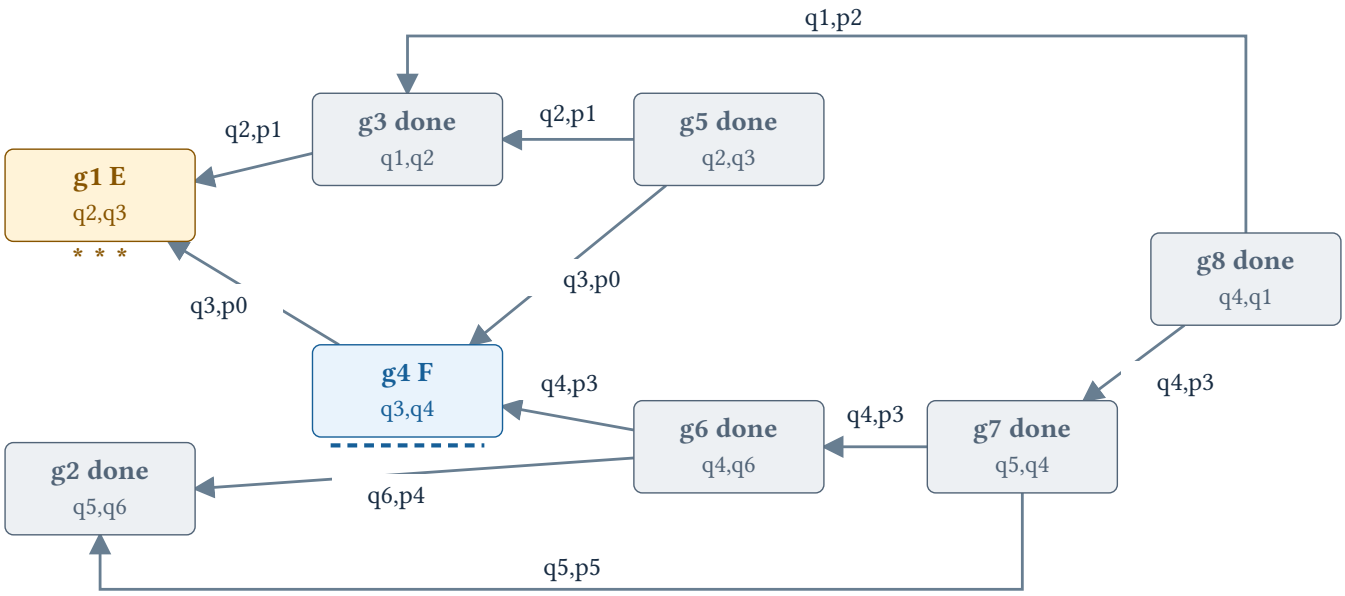
Backward state B3

After B3: visit g2. g4 is still blocked. Next: B4.

Physical line / current occupants



Dependency graph / reversed arrows



--- F: front *** E: lookahead done: already visited wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	2	1	3	2
q2	1	0	1	2	4	3
q3	2	1	0	3	5	4
q4	1	2	3	0	2	1
q5	3	4	5	2	0	1
q6	2	3	4	1	1	0

Layers at this snapshot

F = {g4}

E = {g1}

Gate / physical operands / distance

g4: (p0,p3) d = 3

g1: (p1,p0) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 24: Backward state B3, before B4: g2 is done; F={g4}, E={g1}, and d(q3,q4)=3.

Decision B4. SWAP(p0,p1) ties SWAP(p2,p3) on distance. The ready term prefers p0 and p1 (free at 5, versus 9 for p3). g4 is still blocked, now at distance 2; no gate is visited.

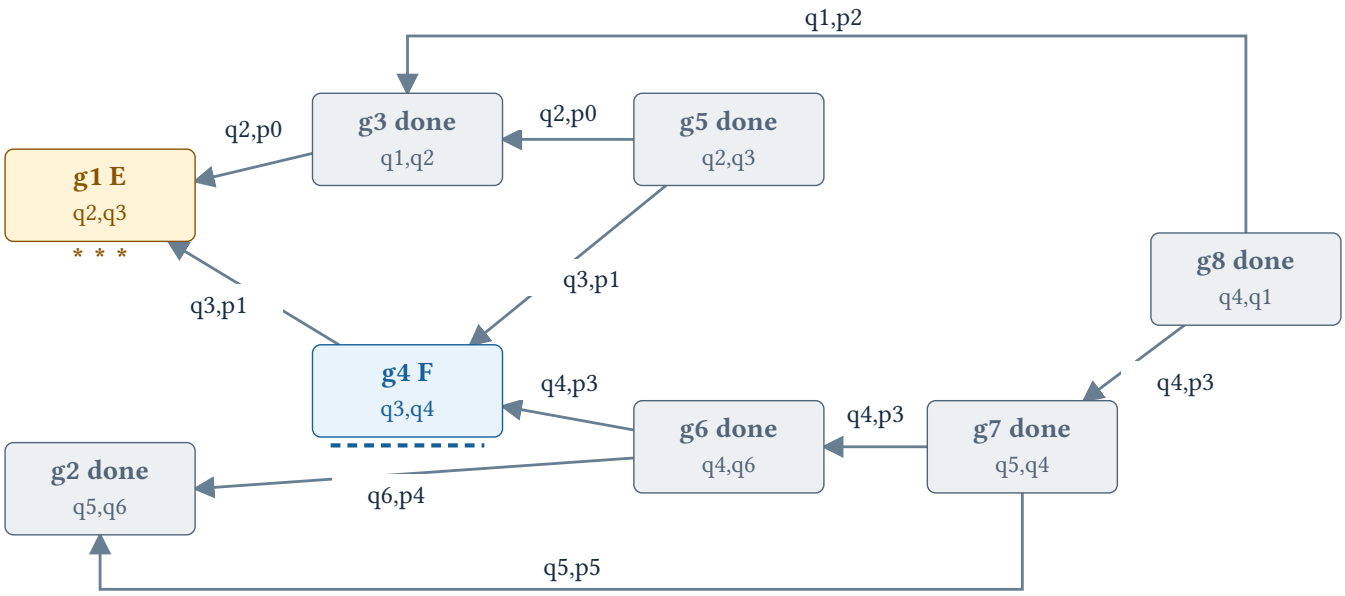
Backward state B4

After B4: no CX visited; g4 is still blocked. Next: B5.

Physical line / current occupants



Dependency graph / reversed arrows



--- F: front *** E: lookahead done: already visited wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	2	1	1	3	2
q2	2	0	1	3	5	4
q3	1	1	0	2	4	3
q4	1	3	2	0	2	1
q5	3	5	4	2	0	1
q6	2	4	3	1	1	0

Layers at this snapshot

F = {g4}

E = {g1}

Gate / physical operands / distance

g4: (p1,p3) d = 2

g1: (p0,p1) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 25: Backward state B4, before B5: q3 moves to p1; F={g4}, E={g1}, and d(q3,q4)=2.

Decision B5. SWAP(p2,p3) makes g4 adjacent. Visit g4 on (p1,p2), then g1 on (p0,p1). The backward pass is complete.

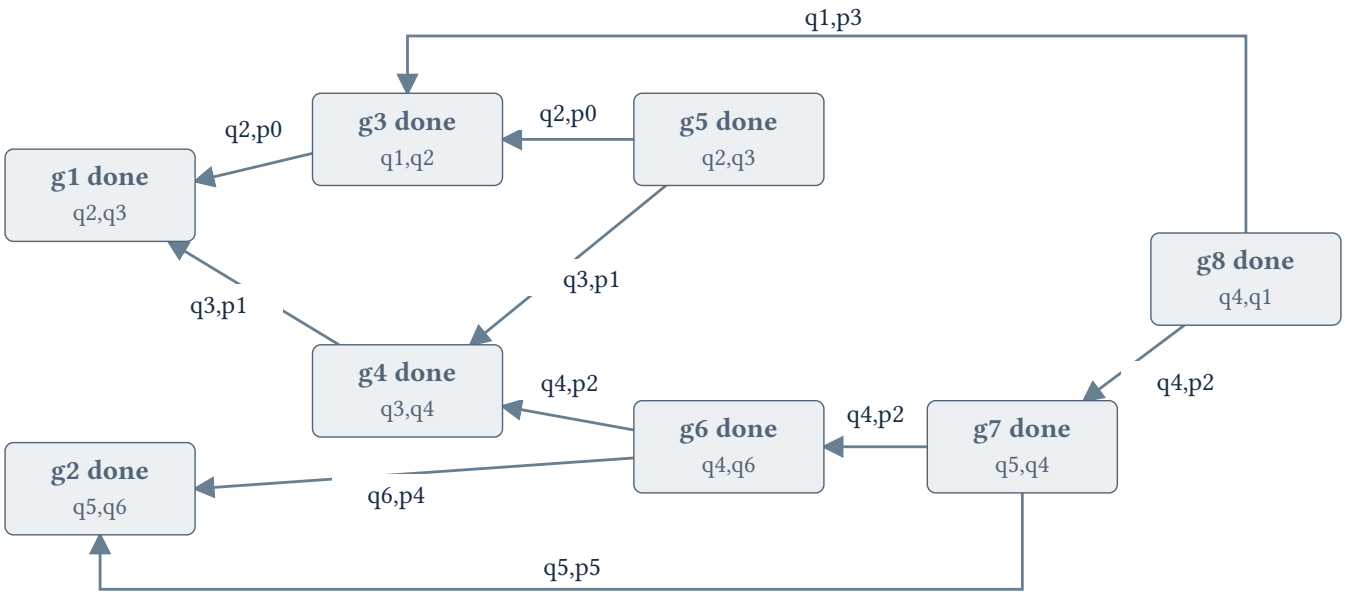
Backward state B5

After B5: visit g4, then g1. Keep this layout; discard the exploratory SWAPs.

Physical line / current occupants



Dependency graph / reversed arrows



--- F: front *** E: lookahead done: already visited wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	3	2	1	2	1
q2	3	0	1	2	5	4
q3	2	1	0	1	4	3
q4	1	2	1	0	3	2
q5	2	5	4	3	0	1
q6	1	4	3	2	1	0

Layers at this snapshot

F = {}

E = {}

Gate / physical operands / distance

No gates remain to route.

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 26: Backward state B5: every gate has been visited. F and E are empty; the resulting layout is q2,q3,q4,q1,q6,q5 on p0 through p5.

The scores behind those choices, showing the winner and its closest rival at each decision (`H_now` is the weighted distance total that decay mode adds back, section 8):

Decision	Candidate	H_now + ΔD_F + w × ΔD_E	Credit	× m	Ready penalty	Total
B1	SWAP(p0, p1)	4.0833 − 1 + 0	−2/3	2.4167	0.000022	**2.416689**
B1	SWAP(p3, p4)	4.0833 − 1 + 0	0	3.0833	0.000022	3.083355
B2	SWAP(p3, p4)	2.0833 − 1 + 0	0	1.0833	0.000022	**1.083355**
B2	SWAP(p4, p5)	2.0833 − 1 + 0.0833	0	1.1667	0	1.166667
B3	SWAP(p3, p4)	6.0833 − 2 + 0	−2/3	3.4167	0.000132	**3.416799**
B3	SWAP(p0, p1)	6.0833 − 1 + 0	0	5.0833	0.000110	5.083443
B4	SWAP(p0, p1)	3.0833 − 1 + 0	0	2.0833	0.000110	**2.083443**
B4	SWAP(p2, p3)	3.0833 − 1 + 0	0	2.0833	0.000198	2.083531
B5	SWAP(p2, p3)	2.0833 − 1 + 0	0	1.0833	0.000198	**1.083531**
B5	SWAP(p1, p2)	2.0833 − 1 + 0.0833	0	1.1667 × 1.001 = 1.1678	0.000176	1.168009

(At B1, SWAP(p1, p2) scores the same as SWAP(p3, p4). At B5 the multiplier 1.001 applies because SWAP(p0, p1) was chosen at B4 in the same search.)

The backward pass ends at `[q2, q3, q4, q1, q6, q5]`. Its five SWAPs are discarded, like the six of the forward pass; only this layout survives.

Refined forward pass: states R0 to R4

Use B5′s layout as the initial layout and route the real circuit forward, still with the decay configuration, since `SabreLayout` does the final routing too:

Start, R0. Reset progress for a new forward pass. One-qubit gates run as soon as their dependencies allow. g1 on (p0,p1), g2 on (p5,p4), and g4 on (p1,p2) are adjacent and are emitted before any SWAP. `F = {g3,g6}`: their distances are 3 and 2. `E = {g5,g7}`.

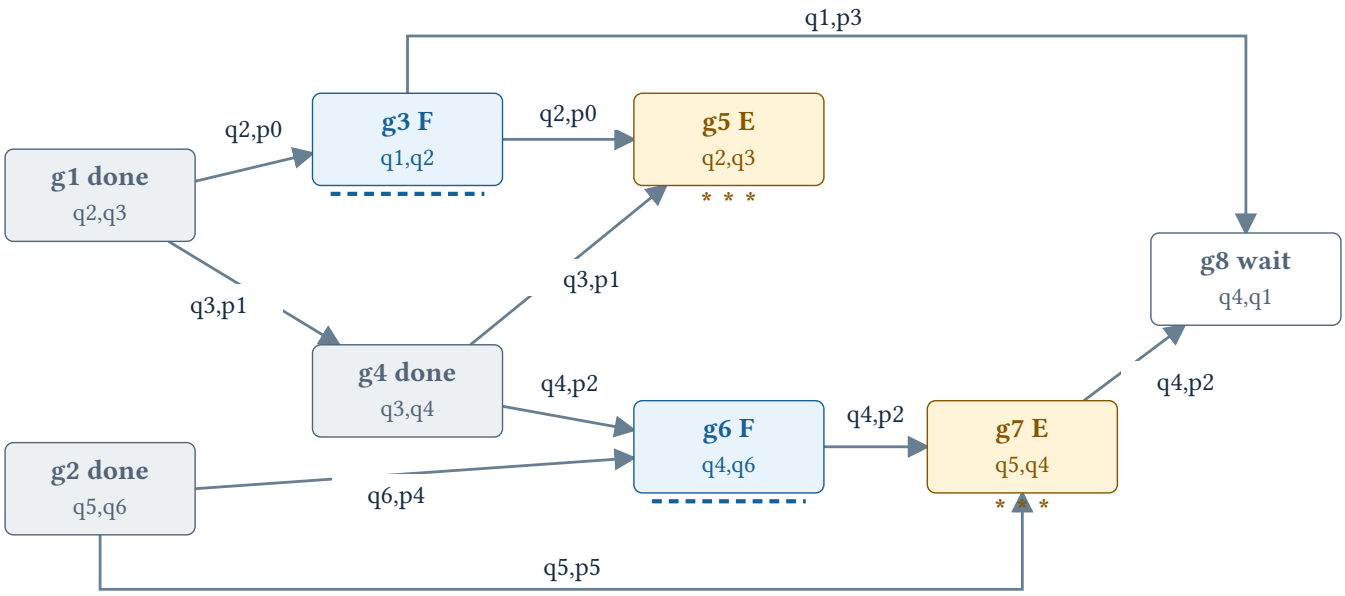
Refined forward state R0

Before R1: emit g1, g2 and g4 without a SWAP.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	3	2	1	2	1
q2	3	0	1	2	5	4
q3	2	1	0	1	4	3
q4	1	2	1	0	3	2
q5	2	5	4	3	0	1
q6	1	4	3	2	1	0

Layers at this snapshot

F = {g3, g6}

E = {g5, g7}

Gate / physical operands / distance

g3: (p3,p0) d = 3

g6: (p2,p4) d = 2

g5: (p0,p1) d = 1

g7: (p5,p2) d = 3

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 27: Refined forward state R0, before R1: g1, g2, and g4 are done. Forward arrows are restored; F={g3,g6}, E={g5,g7}.

Decision R1. SWAP(p2,p3) has `ΔD_F = -2`, helping g3 and g6, and also brings g7′s pair closer. Emit g6 on (p3,p4), with its Z. g7 joins g3 in the front. Both g5 and g8 are in lookahead: pretending the entire front is done satisfies their remaining dependencies.

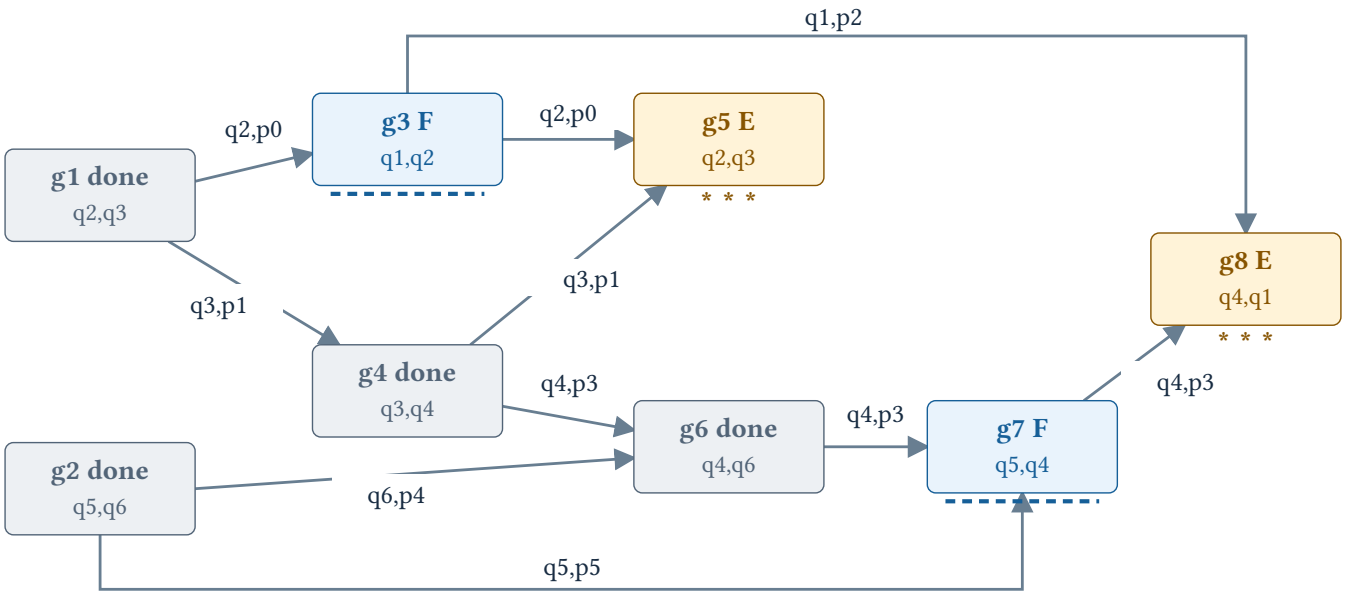
Refined forward state R1

After R1: emit g6. g7 joins g3 in the front. Next: R2.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	2	1	1	3	2
q2	2	0	1	3	5	4
q3	1	1	0	2	4	3
q4	1	3	2	0	2	1
q5	3	5	4	2	0	1
q6	2	4	3	1	1	0

Layers at this snapshot

F = {g3, g7}

E = {g5, g8}

Gate / physical operands / distance

g3: (p2,p0) d = 2

g7: (p5,p3) d = 2

g5: (p0,p1) d = 1

g8: (p3,p2) d = 1

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 28: Refined forward state R1, before R2: g6 is done, F={g3,g7}, and E={g5,g8}. Both front pairs have distance 2.

Decision R2. SWAP(p3,p4) makes g7 adjacent, with a fusion credit from g6. Emit g7 on (p5,p4). g8 must still wait for g3, so it stays in E.

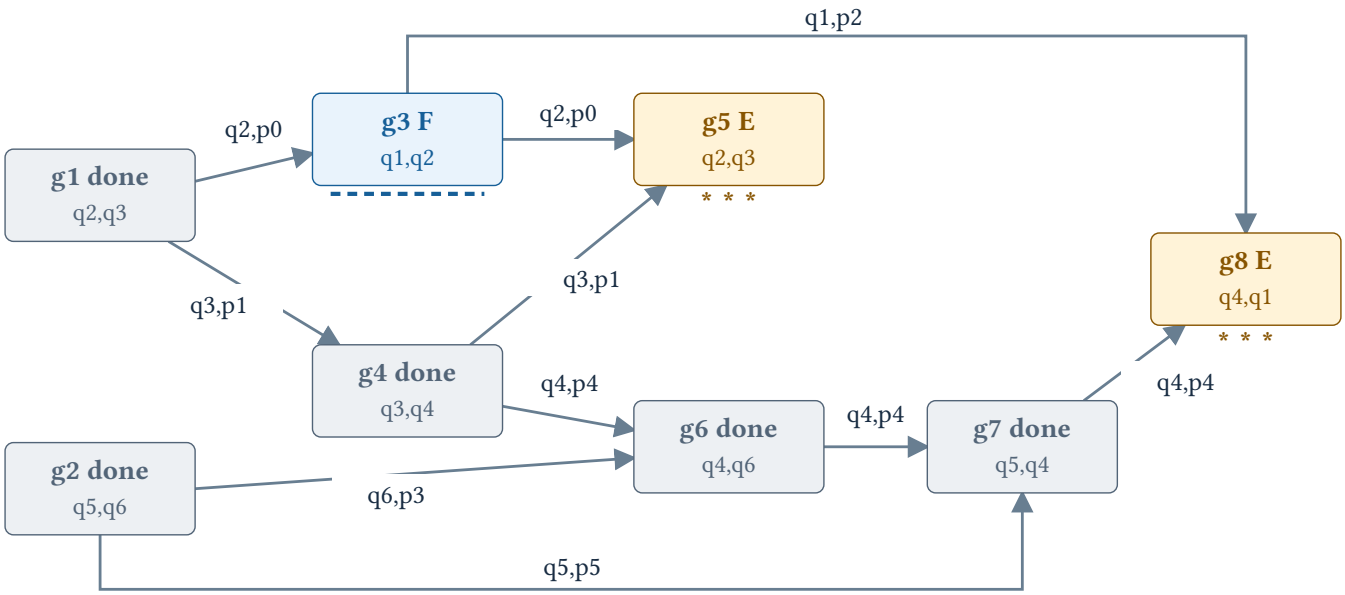
Refined forward state R2

After R2: emit g7. g8 still waits for g3. Next: R3.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	2	1	2	3	1
q2	2	0	1	4	5	3
q3	1	1	0	3	4	2
q4	2	4	3	0	1	1
q5	3	5	4	1	0	2
q6	1	3	2	1	2	0

Layers at this snapshot

F = {g3}

E = {g5, g8}

Gate / physical operands / distance

g3: (p2,p0) d = 2

g5: (p0,p1) d = 1

g8: (p4,p2) d = 2

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 29: Refined forward state R2, before R3: g7 is done; F={g3}, E={g5,g8}. g8 remains in lookahead despite g7's emission because g3 is unfinished.

Decision R3. SWAP(p0,p1) makes g3 adjacent without disturbing g5 or g8, which SWAP(p1,p2) would have pulled apart. Emit g3 on (p2,p1), then g5 on (p1,p0). g8 becomes the front, at distance 2.

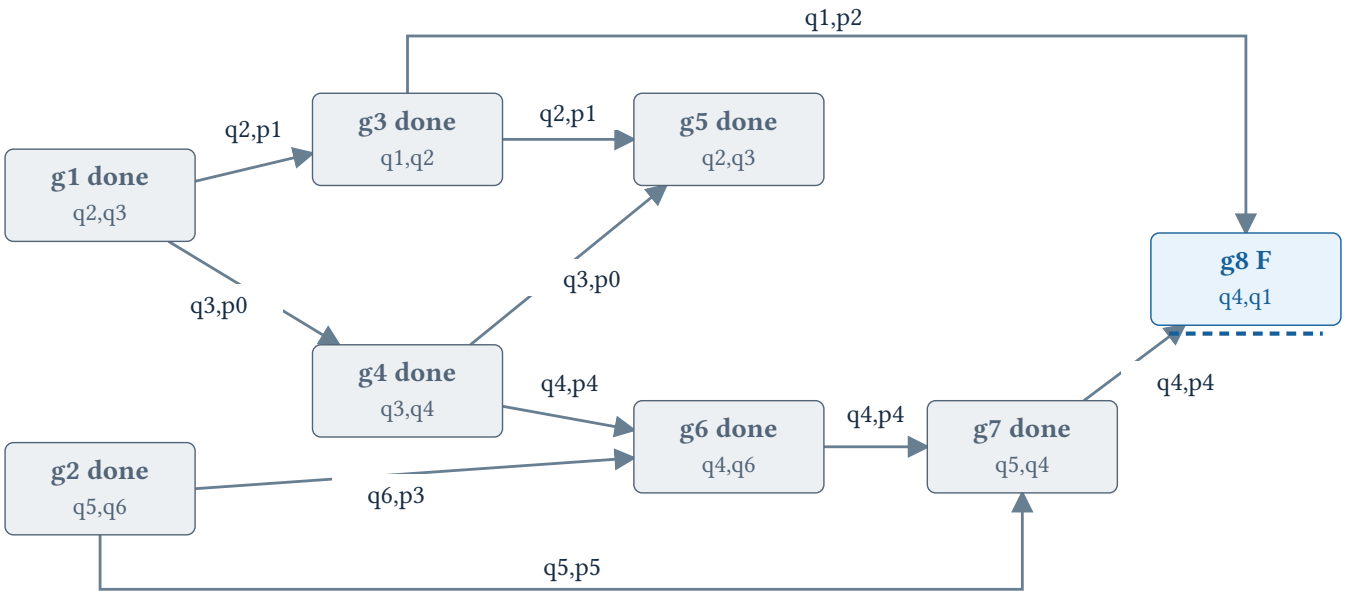
Refined forward state R3

After R3: emit g3, then g5. Next: R4.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	1	2	2	3	1
q2	1	0	1	3	4	2
q3	2	1	0	4	5	3
q4	2	3	4	0	1	1
q5	3	4	5	1	0	2
q6	1	2	3	1	2	0

Layers at this snapshot

F = {g8}

E = {}

Gate / physical operands / distance

g8: (p4,p2) d = 2

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 30: Refined forward state R3, before R4: g3 and g5 are done; F={g8}, E is empty, and d(q4,q1)=2.

Decision R4. SWAP(p2,p3) ties SWAP(p3,p4) on distance. The ready term prefers p2 and p3 (free at 9, versus 10). Emit g8 on (p4,p3). Finished.

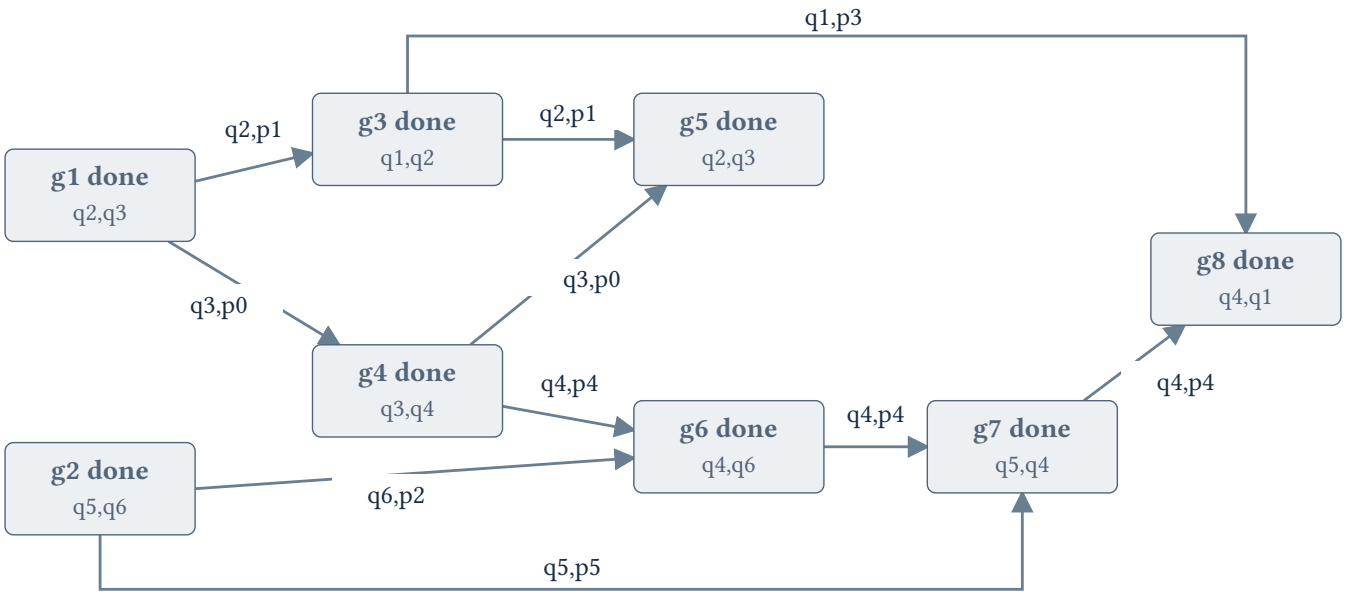
Refined forward state R4

After R4: emit g8. Four SWAPs; the final mapping is shown above.

Physical line / current occupants



Dependency graph / forward arrows



--- F: front *** E: lookahead done: already emitted wait: later gate

Distances between logical qubits

	q1	q2	q3	q4	q5	q6
q1	0	2	3	1	2	1
q2	2	0	1	3	4	1
q3	3	1	0	4	5	2
q4	1	3	4	0	1	2
q5	2	4	5	1	0	3
q6	1	1	2	2	3	0

Layers at this snapshot

F = {}

E = {}

Gate / physical operands / distance

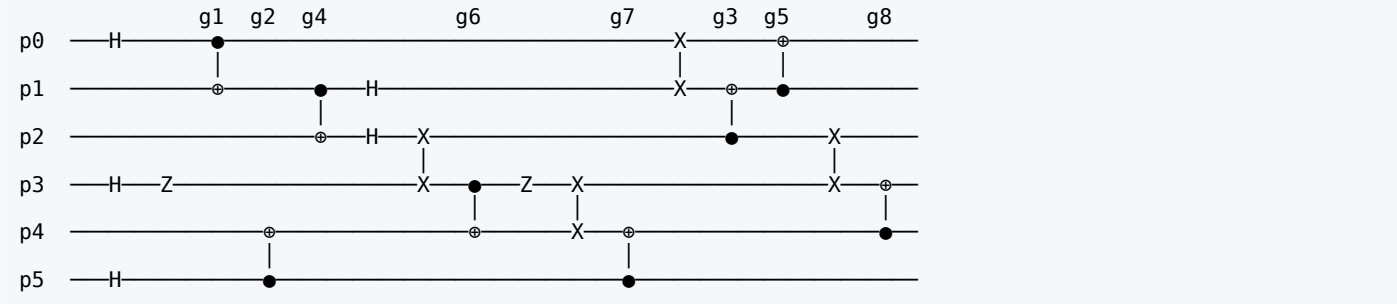
No gates remain to route.

Edge labels give current locations, even beside completed gates. They are not hardware links.

A SWAP changes the matrix; emitting a gate leaves the layout and matrix unchanged.

Figure 31: Refined forward state R4: every gate is done, F and E are empty, and the final layout is q3,q2,q6,q1,q4,q5.

The output circuit:



Four SWAPs instead of six. The eleven exploratory SWAPs used during the search (six forward, five backward) were never added to any output. Only the layout they produced survives. Note also how the final route emits three CX gates before its first SWAP: the layout was shaped by the backward pass to suit the start of the circuit.

Larger circuits benefit from repeating the forward/backward pair several times. Each repetition is a heuristic pass, so improvement is likely but not guaranteed.

17. What one layout trial runs

`layout_trial` in `layout.rs` performs these steps for one starting guess:

1. **Complete the starting layout.** Fill in any unspecified program qubits at random, and invent placeholder program qubits ("ancillas") for every physical position the circuit does not use, so that the layout is a full permutation of all positions. A supplied partial layout only seeds the search; its assignments may move during the exploratory passes.
2. **Route forward once.** Keep the final layout; throw away the SWAPs.
3. **Route backward once**, starting from that layout. Keep its final layout.
4. **Repeat steps 2–3 `max_iterations` times.** The default of 3 means three forward and three backward passes. Each pass starts with fresh decay, ready-time and fusion history; only the layout is carried over.
5. **Route the real circuit** from the resulting layout, `swap_trials` times with different seeds, and keep the attempt with the fewest SWAPs.

The pass runs many such trials from different starting guesses and keeps the one whose final routing used the fewest SWAPs:

Starting guesses tried	Count
Fully random layouts	`layout_trials` (default: Qiskit's default process count)
User-supplied partial layouts	as many as given
A densely connected subset of the device (`dense_layout`)	1
Program qubit i at position i	1
Program qubit i at position $n - 1 - i$	1
A device-specific ring, for 127-, 133- and 156-qubit IBM devices	1 if applicable

Three parameters, often confused:

Parameter	Controls
`max_iterations`	Forward/backward pairs per starting guess.
`layout_trials`	Number of random starting guesses.
`swap_trials`	Final routing attempts per refined layout. The exploratory passes always run one attempt each.

All of these are classical search effort. None of them run anything on a quantum device.

One subtlety of the exploratory passes. They route a stripped-down graph (`only_interactions`) that keeps the two-qubit pairs, dependencies and `block` values but drops the original operations. Two consequences: these passes cannot rebuild a circuit (which is fine, their output is discarded), and a barrier or measurement that forms its own node can no longer be inspected, so it does **not** clear fusion history during the exploratory passes the way it does in final routing. The fusion credit can therefore be slightly more generous while the layout is being searched than when the output is routed.

Part V — The rest of the picture

18. How the circuit is compressed before routing

The router does not look at the original circuit gate by gate. `SabreDAG::from_dag` first builds a smaller graph of **routing nodes**, folding together operations that cannot change any routing decision:

- One-qubit gates never need routing, so each one is folded into the node that precedes it on its wire.
- Consecutive two-qubit gates on the **same pair** are folded into one node: once the pair is adjacent, all of them can run.

Example: five operations become one "run immediately" operation plus two routing nodes.

Operation	What happens to it
`h(0)`	Nothing precedes it and it needs no routing: it goes into `initial`, the list emitted before routing starts.
`cx(0, 1)`	Becomes routing node R1, a `TwoQ` node for the pair (0, 1).
`rz(0.1, 1)`	One-qubit gate whose only predecessor is R1: folded into R1.
`cx(0, 1)`	Same pair, both wires come from R1: folded into R1. R1's `block` becomes 2.
`cx(0, 2)`	Wire 0 comes from R1 but wire 2 is new: becomes node R2, with a dependency R1 → R2.

The router only ever decides about R1 and R2. When R1 is routed, its three operations are emitted together, in order. This is bookkeeping, not a claim that the three gates are one hardware operation.

In the running example of Part II no two consecutive CX gates share a pair, so each CX is its own routing node, and the one-qubit gates distribute like this: H(q1), Z(q1), H(q2) and H(q5) have no two-qubit gate before them on their wires and go into `initial`; H(q3) and H(q4) are folded into g4's node; Z(q4) is folded into g6's node. That is why section 7's emit sweep writes the two H gates right after g4 and the Z right after g6, and why those one-qubit gates add nothing to the nodes' `block` values.

Each `SabreNode` holds:

Field	Content
`indices`	The original circuit operations, kept so the output can be rebuilt.
`kind`	`TwoQ` (a pair to make adjacent), `Synchronize` (an ordering point such as a barrier or a measurement that needs no routing), or `ControlFlow` (a nested block).
`block`	The fusion estimate of section 11, or none if the node contains a block-ending operation.

Operations on more than two qubits are treated as synchronization points; SABRE expects earlier passes to have decomposed them.

19. Control flow and disconnected devices

Control flow. A circuit can contain an `if` whose body is itself a circuit. `route_control_flow_block` routes each body recursively with its own fresh state, then appends SWAPs that bring the body's final layout back to the layout at its entry. That way the code after the `if` sees the same layout whichever branch ran. Those restoring SWAPs are real output, unlike the discarded search SWAPs of Part IV. Because every block restores its layout, the exploratory layout passes treat control-flow nodes as plain synchronization points.

Example: a branch is entered with `[q1, q2, q3]`, its body needs `CX(q1, q3)`, routes it with SWAP(p0, p1), and ends at `[q2, q1, q3]`. One more SWAP(p0, p1) is appended inside the branch to return to `[q1, q2, q3]`. An empty `else` branch already has that layout.

Disconnected devices. If the device has several unconnected pieces, the layout pass first splits the circuit into independent components and assigns each to a piece that can hold it, then routes. A gate between two qubits on different pieces can never be routed: no sequence of SWAPs crosses a missing connection. At the other extreme, if every pair of positions is directly connected (all-to-all coupling), no routing is ever needed and both passes return early.

20. Map of the Rust code and a reading order

File	Responsibility	Names to look for
mod.rs	Registers the Python bindings.	sabre
dag.rs	Compresses the circuit into routing nodes (section 18); computes block.	SabreDAG::from_dag, SabreNode, InteractionKind, block_gates, only_interactions, reverse_dag
layer.rs	Stores the front and lookahead layers; computes distance changes.	Layers, Layer::score, Layer::total_score
heuristic.rs	The configurable weights, decay settings, tie epsilon and attempt limit.	Heuristic, BasicHeuristic, LookaheadHeuristic, DecayHeuristic, SetScaling
route.rs	The routing loop, scoring, safety valve, control flow, and output rebuild.	swap_map_trial, choose_best_swap, update_route, advance_swap, advance_routed, force_enable_closest_node, swap_map, RoutingResult::rebuild
layout.rs	The forward/backward layout search and the list of starting guesses.	sabre_layout_and_routing, layout_trial, add_heuristic_layouts
vec_map.rs	A typed array: VecMap<PhysicalQubit, u32> cannot be indexed by a program qubit by mistake.	VecMap

The Python entry points are SabreSwap (routing only; the circuit's wires are already physical positions) and SabreLayout (layout search, which by default also routes). SabreSwap builds a Heuristic from the mode name; SabreLayout always uses the decay configuration. Both call into Rust through qiskit._accelerate.sabre.

Suggested reading order. swap_map_trial for the loop; choose_best_swap for the decision; Layer::score for the distance delta; update_route for the emit-and-advance sweep; advance_swap and advance_routed for this checkout's history; then SabreDAG::from_dag and layout_trial. Leave rebuild_onto and control flow for last.

21. Python examples you can run

These need Qiskit built from this checkout, including its Rust extension. The outputs below were produced from such a build. Another Qiskit version may make different seeded choices.

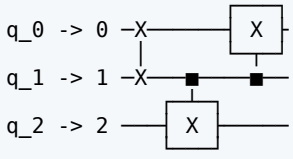
The smallest example (section 3)

```
from qiskit import QuantumCircuit
from qiskit.transpiler import CouplingMap
from qiskit.transpiler.passes import SabreSwap

circuit = QuantumCircuit(3)
circuit.cx(0, 2)  # g1: not adjacent on a line
circuit.cx(0, 1)  # g2: adjacent now, but it must wait for g1

routed = SabreSwap(CouplingMap.from_line(3), heuristic="lookahead", seed=0, trials=1)(circuit)
print(routed.draw(output="text"))
for instruction in routed.data:
    print(instruction.operation.name, [routed.find_bit(q).index for q in instruction.qubits])
```

Output:




```
swap [1, 0]
cx [1, 2]
cx [1, 0]
```

Read the drawing left to right; each wire is a physical position, and q_0 -> 0 says program qubit q_0 starts at position 0. The two X marks joined by a line are the SWAP. Exactly as worked in section 3: SWAP(p0, p1), then both CX gates without further SWAPs. The lookahead term makes this choice unique, so every seed gives the same result.

One gate, a genuine coin flip

```
circuit = QuantumCircuit(3)
circuit.cx(0, 2)

routed = SabreSwap(CouplingMap.from_line(3), heuristic="decay", seed=7, trials=1)(circuit)
print(routed.draw(output="text"))
print(routed.count_ops())
```

Output:

```
q_0 -> 0 ──────■──
              │
q_1 -> 1 ─X──┌─┴─┐─
          │  │ X │
q_2 -> 2 ─X──└───┘─

OrderedDict([('swap', 1), ('cx', 1)])
```

With no second gate there is nothing to look ahead to, every ready time is 0, and no fusible block exists. SWAP(0, 1) and SWAP(1, 2) score identically, and the seeded generator picks one. Over many seeds, both appear.

The running example (Part II)

```
from qiskit import QuantumCircuit
from qiskit.transpiler import CouplingMap
from qiskit.transpiler.passes import SabreSwap

def program():
    """The figure's circuit. Wire i holds program qubit q(i+1)."""
    qc = QuantumCircuit(6)
    q = lambda i: i - 1
    qc.h(q(1)); qc.h(q(2)); qc.h(q(5))
    qc.cx(q(2), q(3))  # g1
    qc.cx(q(5), q(6))  # g2
    qc.z(q(1))
    qc.cx(q(1), q(2))  # g3
    qc.cx(q(3), q(4))  # g4
    qc.h(q(3)); qc.h(q(4))
    qc.cx(q(2), q(3))  # g5
    qc.cx(q(4), q(6))  # g6
    qc.z(q(4))
    qc.cx(q(5), q(4))  # g7
    qc.cx(q(4), q(1))  # g8
    return qc

circuit = program()
hardware = CouplingMap.from_line(6)

# SabreSwap treats the circuit's wires as physical positions. To start from
# the occupants [q1, q2, q6, q3, q4, q5], rebuild the circuit with each
# program qubit on the wire of its starting position.
occupants = [1, 2, 6, 3, 4, 5]
position = {q: p for p, q in enumerate(occupants)}
placed = QuantumCircuit(6)
for inst in circuit.data:
    wires = [position[circuit.find_bit(b).index + 1] for b in inst.qubits]
    placed.append(inst.operation, wires)

routed = SabreSwap(hardware, heuristic="lookahead", seed=0, trials=1)(placed)
print("SWAPs:", routed.count_ops().get("swap", 0))
print("Final wire-to-position:", routed.layout.final_index_layout())
for instruction in routed.data:
    print(instruction.operation.name, [routed.find_bit(q).index for q in instruction.qubits])
```

Output:

```
SWAPs: 6
Final wire-to-position: [2, 0, 4, 1, 3, 5]
h [0]
z [0]
h [1]
swap [3, 2]
cx [1, 2]
cx [0, 1]
swap [4, 3]
cx [2, 3]
h [2]
cx [1, 2]
swap [1, 0]
swap [2, 1]
h [3]
h [5]
cx [5, 4]
cx [3, 4]
z [3]
swap [4, 3]
cx [5, 4]
swap [3, 4]
cx [3, 2]
```

`Final wire-to-position` lists, for each input wire, where its contents end up. Wire 0 started with q1 and ends at position 2, wire 1 started with q2 and ends at 0, wire 2 started with q6 and ends at 4, and so on: that is `[q2, q3, q1, q4, q6, q5]`, the final layout of section 7. (This array is the inverse view of the occupant rows used in the text, which list program qubits by position.)

Two things look different from the hand-worked tables and are not:

- The listing is one topological order of the output, and independent operations on disjoint wires can be listed in any order. For instance `swap [1, 0]` and `swap [2, 1]` (decisions 4 and 5) are printed before `cx [5, 4]` (g2) and `cx [3, 4]` (g6), although section 7 emitted g2 and g6 long before those SWAPs. The dependencies are identical.
- A SWAP's two wires may be printed in either order: `swap [3, 2]` is SWAP(p2, p3).

Section 7 made no random choice, so every seed produces this same output circuit.

Let SABRE choose the initial layout (Part IV)

`SabreLayout` normally starts from random guesses plus a few built-in ones (section 17). To reproduce section 16, hand it the Part II layout as its starting guess through the `sabre_starting_layouts` property, and ask for no random guesses:

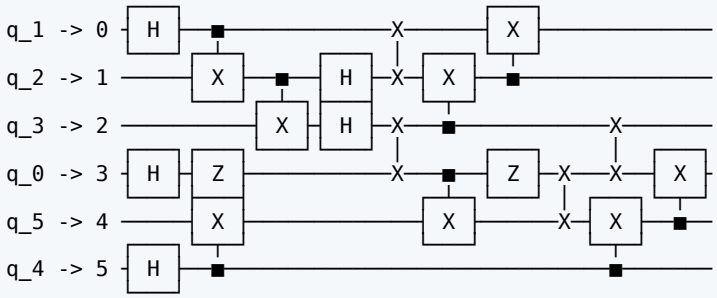
```
from qiskit.transpiler import Layout, PassManager
from qiskit.transpiler.basepasses import AnalysisPass
from qiskit.transpiler.passes import SabreLayout

class StartFrom(AnalysisPass):
    """Hand SabreLayout the starting guess worked in section 16."""
    def run(self, dag):
        self.property_set["sabre_starting_layouts"] = [
            Layout({circuit.qubits[q - 1]: p for p, q in enumerate(occupants)})
        ]

compiled = PassManager([
    StartFrom(),
    SabreLayout(hardware, seed=0, max_iterations=1, layout_trials=0, swap_trials=1),
]).run(circuit)
print("SWAPs:", compiled.count_ops().get("swap", 0))
print("Initial:", compiled.layout.initial_index_layout(filter_ancillas=True))
print("Final:", compiled.layout.final_index_layout())
print(compiled.draw(output="text"))
```

Output:

```
SWAPs: 4
Initial: [3, 0, 1, 2, 5, 4]
Final: [3, 1, 0, 4, 5, 2]
```



`Initial: [3, 0, 1, 2, 5, 4]` lists the starting position of q1 … q6: q1 at p3, q2 at p0, q3 at p1, q4 at p2, q5 at p5, q6 at p4, which is the occupant row `[q2, q3, q4, q1, q6, q5]` found by the backward pass in section 16. `Final: [3, 1, 0, 4, 5, 2]` is `[q3, q2, q6, q1, q4, q5]`, where section 16′s final route ended. `layout_trials=0` asks for no random guesses, but the pass still tries its built-in ones; the hand-seeded trial comes first in the list, and among trials with equal SWAP counts the earliest wins, so this result is the trial worked by hand. (With the defaults, `layout_trials=1` and no seed layout, the pass also finds a four-SWAP route, from the built-in guess that places q1 … q6 in reverse order.)

The output still contains high-level `swap` operations. Translating them to the device's native gates and merging them with neighbors is done by later passes.

22. Common confusions

"Does a SWAP physically move the qubits?" No. The positions are fixed hardware. A SWAP exchanges the *states* held by two positions; the layout records where each program qubit's state now lives.

"Why is a good score negative?" Scores are *changes* in summed distance. Negative means the SWAP brings pairs closer. It is not a count of anything.

"Why isn't the front layer just all ready gates?" Ready gates that are also adjacent are emitted on the spot, so they never wait in the front layer. The front layer holds the gates that need a SWAP.

"Why are g3 and g4 in the lookahead layer at the start, when they are already adjacent?" So that the score can notice a SWAP that would pull them apart. Being adjacent now is information worth protecting. (Decision 1 does pull g4 apart, and pays for it in the lookahead term.)

"Does SABRE consider every possible SWAP?" No. Only hardware edges that touch a front-layer qubit are candidates. Interior edges can still be reached, one decision at a time, as the front qubits move.

"Is the result optimal?" No. SABRE makes one cheap, informed choice at a time and never looks back further than the safety valve's rollback. It is fast and usually good, which is the point of a heuristic.

"Is decay about decoherence?" No. It is a small multiplier that discourages reusing the same wires in consecutive search SWAPs.

"Why does my printed circuit order differ from the walkthrough?" Independent operations on disjoint positions can be listed in any order. Only the dependencies are fixed.

"Does the final layout have to be undone?" Not at the top level. Qiskit records the final layout, and measurements are placed on the correct positions. Inside control-flow blocks, the layout *is* restored, with real SWAPs.

"Are the search SWAPs of the layout pass in my circuit?" No. Forward and backward passes only produce a layout; their SWAPs are thrown away. Only the final routing's SWAPs appear in the output.

"Does `heuristic="basic"` turn off fusion and ready-time scoring?" No. In this checkout all three modes include them. `basic` only removes lookahead and decay and averages the distance term over the front gates.

23. Check your understanding

Try to answer before reading the explanation.

1. g3′s qubits are adjacent at the start. Why not emit it first?

Because g3 depends on g1, which has not been emitted. Adjacency is one of two conditions; readiness is the other. Lookahead may score g3′s distance, but that is a prediction, not permission to run it.

2. On `p0 — p1 — p2 — p3 — p4`, the only front gate uses p0 and p4. Is `(p1, p2)` a candidate?

No. Neither endpoint holds a front qubit. The candidates are `(p0, p1)` and `(p3, p4)`. After one of them is applied, the front qubit has moved, and the candidate list is rebuilt around its new position.

3. Why not always move one qubit along the shortest path to its partner?

Because that ignores every other pending and upcoming gate. In section 6, SWAP(p1, p2) would have enabled g1 by itself but pushed q6 one step further from q5; scoring the whole front found SWAP(p2, p3), which serves both front gates. The shortest-path move is reserved for the safety valve.

4. The route of section 7 contains SWAP(p3, p4) three times. Do any two of them cancel?

No. Between the first and the second, g2, g4, g6 and g5 are emitted, and g6 uses exactly the wires p3 and p4. Between the second and the third, g7 uses p4. Removing a pair of SWAPs while leaving those gates on the same physical positions would change the computation.

5. The largest ready time in section 12 is 16. Is the circuit's two-qubit depth 16?

Not necessarily. The ready time is an estimate under a fixed accounting rule (SWAP = 3 layers, routed node = 1). Later synthesis, fusion, barriers and control flow can all make the real depth differ.

6. Why does the layout pass route backward if the circuit runs forward?

To carry what the end of the circuit needs back to the start. The forward pass adapts the layout to the late gates; the backward pass adapts that layout to the early gates. Only the resulting layout is kept; the output is then routed forward from it.

7. Can a route with fewer SWAPs still have more native gates or more depth?

Yes. Fusion makes some SWAPs cheaper than others, and parallel SWAPs cost less depth than sequential ones. Complete attempts are still compared by SWAP count, so the chosen attempt is not guaranteed to win on those other measures.

8. What would classic SABRE do at decisions 4, 5 and 6 of section 12?

Flip a coin at each of them, between two SWAPs that close the distance equally. Some outcomes put consecutive SWAPs on the same busy wires, making them sequential. The ready-time term picks the idle pair every time, which is why every seed produces the same circuit in section 21.

24. Glossary

Term	Meaning
Program qubit (logical, virtual)	A qubit name in the user's circuit: q1 … q6 in the running example, A, B, C in Part I.
Physical position (physical qubit)	A numbered place on the device: p0, p1, …
Layout	The mapping from program qubits to physical positions, updated after every SWAP.
Wire	A horizontal line in a circuit drawing; for the output circuit, the same thing as a physical position.
Emit	Write a gate into the output circuit on the positions its qubits occupy at that moment.
Native gate	A gate the device can run directly. Later passes translate everything, including SWAPs, into native gates.
Ancilla	A placeholder program qubit invented to fill a physical position the circuit does not use.
Coupling map	The device's graph of which positions can run a two-qubit gate.
Distance	Fewest edges between two positions. A pair needs distance 1 to interact.
Ready	All of a gate's predecessors have been emitted.
Adjacent	A gate's two qubits are on connected positions.
Front layer `F`	Gates that are ready but not adjacent.
Lookahead layer `E`	Gates that would become ready once the front layer is done.
Candidate SWAP	A hardware edge touching at least one front-layer qubit.
`ΔD_F`, `ΔD_E`	Change in summed front / lookahead distances caused by a candidate.
Decay	A per-position multiplier that discourages reusing recently swapped wires.
Fusion credit	This checkout's estimate of native gates saved when a SWAP merges into the preceding block on the same pair.
Ready time	This checkout's estimate of when a position's assigned two-qubit work ends.
Safety valve	Rollback plus shortest-path SWAPs when a search makes no progress.
Trial	One complete routing attempt with one seed. The attempt with fewest SWAPs wins.
Forward / backward pass	Routing the dependency graph in its normal or reversed direction to refine the initial layout.
Heuristic	A cheap rule for ranking choices that usually works well but offers no optimality guarantee.